

Can Foundation Models Really Understand Economic Research? A Pipeline-Level Benchmark for Agentic Research Systems

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Submission Id: «91»

Abstract

Agentic systems are beginning to automate economic research end-to-end, yet we lack a systematic account of where the underlying model layer fails. Existing benchmarks probe isolated capabilities rather than the research workflow itself. We introduce AER-bench—the Agentic Economic Research benchmark—which evaluates foundation models across the full economic-research pipeline. Its seven task types are extracted from the core stages of the agentic economic-research workflow: Economic Knowledge (EK), Literature Review (LR), Concept Identification (CI), Model Generation (MG), Mathematical Modeling (MM), Economic Analysis (EA), and Paper Abstraction (PA). Built from 191 articles in top economics journals and 900 expert-annotated pre-1950 text segments, AER-bench comprises 1,629 items scored by a hybrid framework combining rule-based metrics and two independent LLM judges. Evaluating 16 frontier models, we find that (1) their own economic knowledge remains volatile and unreliable, pinpointing where research agents require external retrieval; (2) model families exhibit distinct capability fingerprints; and (3) their grasp of historical economic concepts is uneven, exposing gaps in their command of the history of economic thought. AER-bench provides the missing evaluation layer for the emerging field of agentic economic research; code and data are available at GitHub.

Keywords

Benchmarks, LLM evaluation, Agentic AI, Economic research automation, Foundation models

ACM Reference Format:

Anonymous Author(s). 2027. Can Foundation Models Really Understand Economic Research? A Pipeline-Level Benchmark for Agentic Research Systems. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, 3–7 May 2027, IFAAMAS, 39 pages.

1 Introduction

Economic research is a primary input to policy, market design, and aggregate welfare [1, 3, 6, 7, 9, 13, 22], yet producing each contribution still demands a labor-intensive pipeline: literature review, data collection and cleaning, specification of theoretical or econometric models, derivation or estimation, robustness checking, and result interpretation. A substantial share of this pipeline is mechanical and repetitive, and a growing literature has begun to ask how generative AI could absorb it [10].



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Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027), M. Baldoni, F. Fang, W. Yeoh, N. Yorke-Smith (eds.), 3–7 May 2027, Hanoi, Vietnam. © 2027 International Foundation for Autonomous Agents and Multiagent Systems (www.ifaamas.org).

Table 1: Comparison of AER-bench with existing economic scenario benchmarks. AER-bench is the first benchmark designed to evaluate the pipeline of agentic economic research.

Benchmark	Data Source	Focus	# Items	# Tasks
EconGym [14]	Real-world data	Behavior	—	25+
GDPVal [18]	Reports	Productivity	1,320	9
EconLogicQA [19]	News	Reasoning	650	1
FinMR [5]	Web + Experts	Reasoning	3,700	15
AER-bench (Ours)	Journals	Research	1,629	7

Frontier foundation models are now plausible candidates for that role. Web-scale pretraining has given them broad command of economic terminology, canonical models, and econometric techniques [2, 23], and progress on multi-step reasoning, quantitative problem solving, and formal mathematics [8, 11, 25] brings the symbolic and computational manipulations central to economic analysis within reach. Building on foundational paradigms such as ReAct and Toolformer [20, 26], agentic frameworks like OpenClaw [17], Claude Code [4], and Hermes [15] now package these capabilities into reusable, tool-using workflows that browse, execute code, and iterate on intermediate artifacts, while analogous systems have begun to drive scientific and algorithmic discovery end-to-end [12, 16]. A first attempt has already appeared in economics itself: APE (Automatic Policy Evaluation) [21], built on Claude Code, automates empirical policy research, yet on its own self-reported scale still trails top journals by a wide margin—at once demonstrating the promise of agentic economic research and underscoring the need for an independent, rigorous benchmark.

General-purpose academic benchmarks such as MMLU-Pro [24] include an economics subject, but each subject contributes only a few hundred multiple-choice items and cannot probe the research-grade capabilities we are interested in. Several economics-focused benchmarks¹ have therefore been proposed (Table 1). EconGym [14] is a scalable multi-agent simulation testbed in which models play the role of households, firms, or governments across more than twenty real-data-driven economic tasks, evaluating decision behavior rather than research output. GDPVal [18] curates 1,320 tasks across nine GDP-relevant industries from industry reports to measure whether AI can substitute for economically valuable human labor, again outside the academic pipeline. EconLogicQA [19] distills 650 multi-event sequencing questions from economic news articles, targeting the narrow capability of causal and temporal reasoning. FinMR [5] provides over 3,700 multimodal questions on financial-analyst-level chart and formula interpretation, likewise sourced from non-academic web material rather than economics research.

¹We discuss only a few representative economics-related benchmarks here; additional benchmarks covering economic and financial scenarios exist but are omitted for brevity.

Each benchmark is valuable within its own scope—behavioral simulation, productive substitution, event reasoning, and financial analysis, respectively—yet none draws its data from top-journal research articles, and consequently none evaluates the core capabilities that economic research relies on, such as domain knowledge, mathematical modeling, and economic analysis of empirical or theoretical results. This gap remains unaddressed.

To address this gap, we build a benchmark that is *workflow-aligned*: an agentic research system is, at bottom, an orchestration of the stages of the economic-research pipeline around a foundation-model core, and each stage places qualitatively different demands on that core. We therefore decompose the research workflow into its intellectual core stages and instantiate each as a measurable task—Economic Knowledge (EK), Literature Review (LR), Concept Identification (CI), Model Generation (MG), Mathematical Modeling (MM), Economic Analysis (EA), and Paper Abstraction (PA) (Figure 1). Measuring these stages separately yields module-level diagnostics: it tells system builders not merely how good a model is, but *which* stages of an agentic pipeline a model can carry on its own and *which* require external augmentation. As we explain in §2.4, we deliberately evaluate the model layer without external tools—web search is disabled to isolate parametric knowledge and to protect the benchmark from data contamination, and code execution is out of scope because it forms the mechanical periphery of the workflow rather than its intellectual core.

Concretely, we introduce **AER-bench**—the Agentic Economic Research benchmark—the first comprehensive benchmark for systematically and objectively evaluating foundation models on agentic economic research. AER-bench combines two complementary data sources: (i) 191 articles drawn from top-5 economics journals—the *American Economic Review* and *Econometrica*—representing the modern frontier of the field; and (ii) roughly 900 text segments from pre-1950 economics literature, manually annotated by economics doctoral students into 160 concept-identification tasks across eight conceptual dimensions, which support the analysis of how models handle historical economic discourse. Tasks are constructed through two complementary pipelines (§2.2): an LLM-driven multi-round extraction pipeline that turns paper PDFs into self-contained items, and the manual annotation effort just described. The result is a benchmark of 1,629 evaluation items organized into seven task types—spanning capabilities from foundational knowledge recall and symbolic manipulation to higher-order modeling and analysis (§2.1). To score these heterogeneous tasks reliably, AER-bench adopts a hybrid evaluation framework (§2.3): rule-based scoring for EK and CI, and multi-dimensional rubric-based LLM-as-Judge scoring averaged over two independent judges (Claude-4.6-sonnet and gpt-5.2) for the remaining five tasks. We use AER-bench to evaluate 16 frontier models (§3); the full ranking and per-task breakdowns are reported in §4.

In summary, AER-bench makes three contributions. First, it provides a workflow-aligned task taxonomy grounded in top-journal and pre-1950 sources, the first benchmark specifically designed for the emerging field of agentic economic research. Second, its construction methodology—an LLM-driven multi-round extraction pipeline coupled with expert annotation—is itself a reusable asset for extending the benchmark, constructing training data, and adapting the approach to other research disciplines. Third, our

evaluation of 16 frontier models establishes module-level baselines and surfaces pervasive shortcomings: volatile and unreliable parametric economic knowledge, consistent family-level capability fingerprints, and an uneven grasp of historical economic concepts, which we examine in detail in §4. We deliberately evaluate the model layer rather than full agentic systems: the tool-free baselines reported here are the reference points against which complete agentic systems can be measured on the same tasks in future work.

2 Benchmark Design

2.1 Task Taxonomy

To systematically evaluate foundation models on agentic economic research, AER-bench deconstructs the economic-research workflow into seven task types. The taxonomy is not arbitrary: it is derived by extracting the intellectual core of the workflow—the stages a research project must traverse from a source corpus to a finished analysis. A canonical project draws on command of domain knowledge, positions itself in the literature, grounds its concepts in prior work, builds formal models, analyzes empirical or theoretical results, and synthesizes written output. Each stage demands qualitatively different capabilities of the underlying model—factual knowledge, retrieval-like memory, conceptual grounding, symbolic reasoning, and compositional writing—and an agentic research system is, at bottom, an orchestration of exactly these stages around a foundation-model core. AER-bench therefore instantiates each stage as a measurable task:

- **Economic Knowledge (EK):** A rigorous QA task comprising single-choice, true/false, fill-in-the-blank, and ordering questions. It tests the model’s deep comprehension of paper-specific methodologies, findings, and institutional contexts.
- **Literature Review (LR):** Assesses the model’s ability to reconstruct the argument structure of an introduction and appropriately cite prior literature. It is evaluated via a dual mechanism: citation recall (capturing the core reference network) and a paper-specific writing rubric.
- **Concept Identification (CI):** A robust binary classification task requiring models to map text chunks to eight structural economic properties: relevance, mathematical formalization, uniqueness, stability, existence, equilibrium scene, time characteristic, and time horizon.
- **Model Generation (MG):** Evaluates formal construction of an economic model from a de-symbolized narrative. Scoring uses a granular element rubric checking for presence and correctness of sectors, objective functions, constraints, and equilibrium concepts.
- **Mathematical Modeling (MM):** Given a textual economic scenario and notation constraints, the model must perform formal derivations. Outputs are judged on four dimensions: correctness, notation consistency, derivation completeness, and economic intuition.
- **Economic Analysis (EA):** Given uninterpreted empirical or theoretical results, the model generates a qualitative analysis. It is scored across six dimensions: accuracy, economic interpretation, mechanism, theory connection, policy implication, and writing quality.

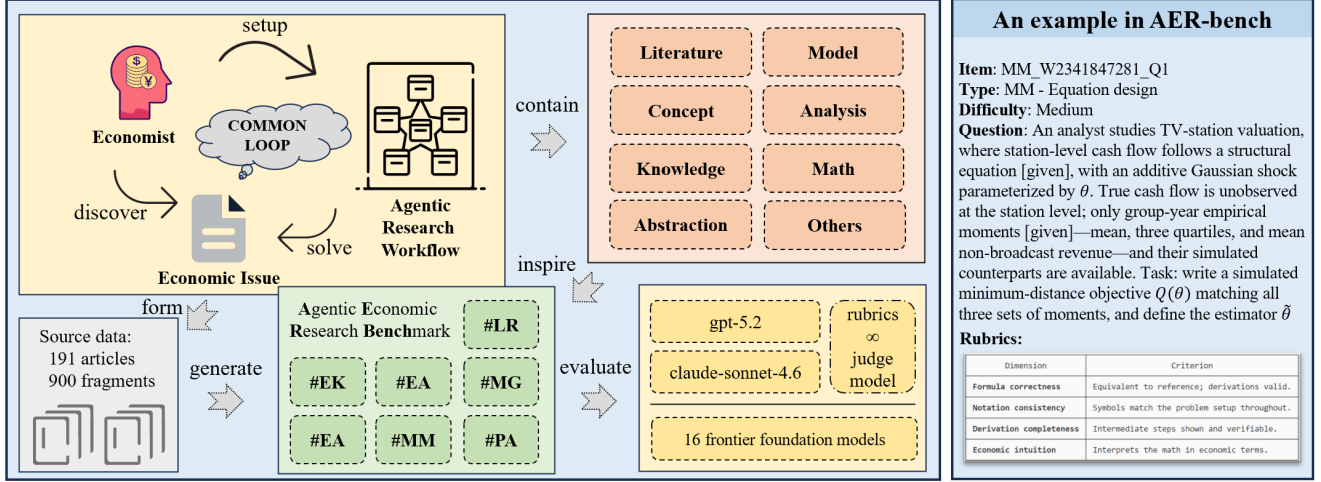


Figure 1: Overview of AER-bench. Left: the overall generation and evaluation framework. Economists study economic issues through agentic research workflows; we then collect a corpus of 191 top-journal articles and 900 historical segments, and parse it to reverse-engineer the seven key components of such workflows. These components inspire the seven tasks of our bench, whose items are generated from the same corpus. Evaluation combines two LLM judges (*claude-4.6-sonnet* and *gpt-5.2*) with a shared rubric and 16 frontier foundation models. Right: a representative task example with a question and its rubric.

- **Paper Abstraction (PA):** The model must synthesize a concise abstract from a condensed paper body, evaluated across 4 dimensions: coverage of atomic semantic propositions, factual accuracy, conciseness, and academic style.

Measuring the stages separately yields module-level diagnostics for building research agents—the intended use of the benchmark—while the tool-dependent periphery of the workflow (search, code execution) is deliberately excluded (§2.4).

2.2 Data Collection & Generation Pipeline

Data Sources. To ensure comprehensiveness, AER-bench combines an automated extraction pipeline for modern research with meticulous manual annotations for historical literature. The dataset is built upon two distinct corpora:

- **Modern Frontiers (191 articles):** Sourced from public, Open Access² versions of papers published in 2025 in top-tier journals (*American Economic Review*, *Econometrica*), guaranteeing that the benchmark challenges models with the most recent methodological innovations while strictly adhering to copyright and open-science protocols.
- **Historical Fragments (900 segments):** Curated from multilingual text segments of canonical historical literature (1900–1950) to evaluate models on historical concept drift and early economic paradigms, which presents a challenging task due to archaic terminology and evolving theoretical frameworks.

Task Generation Pipeline. For the modern corpus, we employ a multi-round, LLM-assisted generation pipeline that begins with structural PDF parsing to extract raw text and equations, followed

by multiple rounds of information extraction, desymbolization, and task construction with automated quality checks to ensure self-contained tasks free of structural errors³. Figure 2 summarizes the round-by-round operations across all six modern-corpus pipelines, illustrating which rounds consume the source PDF and where each pipeline terminates; pipeline depth ranges from two rounds (EK, MM, EA) to four rounds (LR), reflecting the differing structural complexity of each task. In contrast, the historical fragments used for the Concept Identification (CI) task were strictly manually annotated by expert annotators⁴ with advanced degrees in economics, who read each fragment and performed binary classification across eight predefined conceptual fields, providing gold-standard labels that form a rigorous testbed for evaluating AI comprehension of early, non-formalized economic concepts.

2.3 Evaluation Framework

AER-bench utilizes a hybrid evaluation framework that balances absolute objectivity with nuanced semantic assessment, adapting its scoring strategy to the nature of each task type.

Rule-Based Scoring. For tasks with deterministic answers (EK and CI), we employ strict rule-based grading. CI uses categorical exact matching, while EK incorporates unidirectional word-boundary regular expressions for fill-in-the-blank items and normalized Kendall’s τ distance for ordering tasks.

²For articles not available under Open Access licenses from journals, we substitute with legally accessible working paper versions from NBER, arXiv, or authors’ personal homepages to ensure data legality.

³See Appendix H for detailed prompt templates, output formats, and scoring methods for each task.

⁴All annotators are economics doctoral students who have passed their qualifying examinations and language requirements. Moreover, we only select and retain data with consistent annotations to ensure the validity of the task answers.

needs PDF

no PDF (LLM only)

not applicable

✓

task end

	EK	MM	PA	LR	MG	EA
R1	Knowledge Point Extraction	Math Content Extraction	Paper Structure Parsing	Reference Extraction	Full Model Extraction	Result & Data Extraction
R2	Question Generation ✓	Problem Design ✓	Condensed Body Generation	Argument Analysis	Scenario Narrative	Analysis Excerpt & Task Build ✓
R3	—	—	Key Point Verification ✓	Condensed Body Generation	Rubric Construction ✓	—
R4	—	—	—	Rubric Construction ✓	—	—

Figure 2: Auto-acquisition pipelines for the six LLM-generated tasks. Each column is a task type and each row is a generation round; cell labels name the per-round operation. Blue cells consume the source PDF, gray cells operate from prior-round outputs alone (LLM-only), and the green check marks each task’s terminal round. Pipeline depth ranges from 2 to 4 rounds.

LLM-as-Judge. For open-ended generative tasks⁵ (MM, PA, LR, MG, EA), we adopt an LLM-as-Judge paradigm. To prevent holistic scoring biases, each judge model evaluates outputs across multiple isolated dimensions with detailed rubrics, producing a weighted dimension-level score per task that captures distinct aspects of quality. To further enhance robustness, we employ multiple judge models in parallel, mitigating individual model biases. Section 4.2 systematically compares the scoring differences between two parallel judges to validate this approach.

Score Aggregation. For each item, dimension scores are first aggregated into a task score via configurable intra-task weights, then scores from multiple judges are arithmetically averaged to produce a consolidated item score. Overall benchmark scores are computed by weighted summation across task types, with specific weight values detailed in Section 3.

2.4 Design Principles: Isolating the Model Layer

AER-bench is designed around one principle: isolate the capability under test. We evaluate foundation models without external tools, by construction rather than by omission.

Why no web search. We deliberately disable search and retrieval for two reasons. First, *isolation*: the benchmark measures what each model can do from parametric knowledge alone, which is precisely the layer on which any research agent is built; only against a tool-free baseline can the value of retrieval or citation-database augmentation later be quantified. Second, *integrity*: our items are constructed from 2025 publications that are directly retrievable online. Allowing search access would let systems recover the source papers—and their answers—directly, turning a capability measurement into a retrieval contest and compromising the benchmark against data contamination.

⁵Note that the Literature Review (LR) task employs both an LLM-as-Judge rubric and a rule-based citation recall score, which are combined with equal weights.

Why no execution environment. We likewise exclude the data-processing and experimentation stages of empirical research—code execution, file handling, and live computational environments. These stages are mechanical and environment-specific rather than analytical: they form the periphery of the research workflow, not its intellectual core, which is what the seven task types capture. Excluding them also keeps the benchmark reproducible and standardized across providers.

Taken together, these choices mean AER-bench measures the analytical and knowledge core of economic research—the part where model capability, rather than infrastructure, is the binding constraint.

3 Experimental Setup

Evaluated Models & Inference Details. We evaluate 16 state-of-the-art foundation models to establish a comprehensive baseline for agentic economic research. The cohort includes the OpenAI GPT family (gpt-5.4, gpt-5.2, gpt-5, gpt-5.1, gpt-5-mini, gpt-4o), the Anthropic Claude series (claude-4.6-opus, claude-4.6-sonnet, claude-3.7-sonnet), and the Google Gemini lineage (gemini-3.1-pro, gemini-3-flash, gemini-2.5-flash). Furthermore, we incorporate leading open-weight and regional models: kimi-k2.5, glm-5.1, doubao-seed-1.6, and qwen3-v1-32b-instruct. Note that all inferences are executed under a zero-shot setting.

Judge Configuration & Score Aggregation. For the hybrid evaluation framework introduced in Section 2.3, we instantiate a programmatic evaluator for deterministic tasks (EK and CI). For the five open-ended generative tasks, we deploy claude-4.6-sonnet and gpt-5.2 as independent LLM-as-Judge models to mitigate individual evaluator bias.

AI Use Disclosure. The task-generation pipelines described in Section 2.2 use gpt-5.4 as the extraction model under human-designed prompts (full prompt templates in Appendix H); claude-4.6-sonnet

Table 2: Dimension-level scoring weights for each task type, together with the number of items per task and its share of the benchmark. Dimensions are weighted according to their importance for evaluating model performance, with weights normalized to sum to 1 within each task. Detailed rubric definitions for each dimension are provided in Appendix F.

Task	EK	LR	CI	MG	MM	EA	PA
# Items	189	187	160	124	399	380	190
Ratio	11.6%	11.5%	9.8%	7.6%	24.5%	23.3%	11.7%
# Dims	4	2	8	Multiple	4	6	4
Weights	Equal	Equal	Equal	Equal	4:2:3:1	4:4:4:3:3:2	4:4:3:1

and gpt-5.2 serve as the evaluation judges described above. All benchmark scores are produced by the evaluated models themselves under zero-shot inference.

To formalize the multi-judge merging strategy outlined previously, let J be the set of judges evaluating model m on task t . The task-level score is computed as the arithmetic mean of the judges’ outputs:

$$S_{m,t} = \frac{1}{|J|} \sum_{j \in J} s_{m,t}^j \quad (1)$$

The overall benchmark score S_m is then calculated as the weighted sum of the task-level scores. In AER-bench, all seven task categories are assigned equal importance ($w_t = 1/7$):

$$S_m = \sum_{t \in T} w_t \cdot S_{m,t}, \quad \sum_{t \in T} w_t = 1 \quad (2)$$

Within each multi-dimensional task, sub-scores are aggregated according to predefined intra-task dimension weights, which reflect the relative importance of specific economic reasoning facets. These exact dimensional configurations are detailed in Table 2.

4 Results

Throughout this section we report model-layer capabilities: consistent with the design principles of Section 2.4, no external tools are involved, so the baselines below are the reference points against which complete agentic systems can later be measured.

4.1 Model Performance

Table 3 presents the overall benchmark ranking across 16 evaluated foundation models. gpt-5.4 achieves the highest overall score at 75.0%, excelling in Mathematical Modeling (92.8%), Model Generation (85.8%), and Economic Analysis (81.6%); however, a paired bootstrap test shows this lead is statistically indistinguishable from claude-4.6-opus (74.0%, $p = 0.099$), placing the two models in a statistical tie at the top. OpenAI models nonetheless exhibit notable version instability: gpt-4o scores only 54.0% with particularly weak Economic Knowledge (25.6%). In contrast, Anthropic’s Claude models demonstrate remarkably stable performance, with claude-4.6-opus (74.0%), claude-3.7-sonnet (73.7%), and claude-4.6-sonnet (73.4%) clustered within 0.6 percentage points, and achieve the highest Concept Identification scores (72.7–73.0%), substantially outperforming gpt-5.4 (58.9%). Google’s gemini-3.1-pro (71.6%) exhibits task-specific dominance,

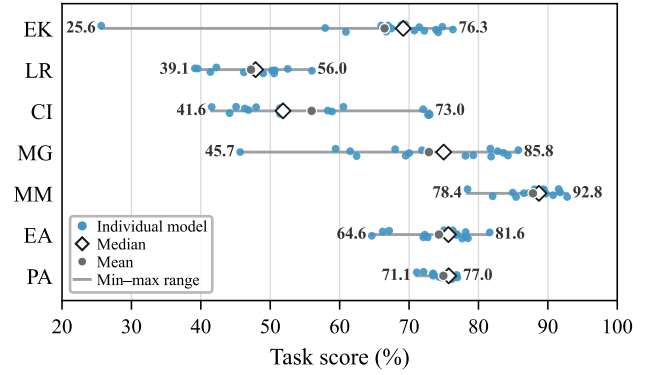


Figure 3: Task score distribution across all 16 evaluated models, one row per task. Blue dots are individual model scores, white diamonds mark the median, gray circles the mean, and the gray bar spans the min-max range with both end-points annotated in bold. The figure reveals substantial performance variation across tasks.

achieving state-of-the-art Economic Knowledge (76.3%) and strong Paper Abstraction (74.3%) despite trailing in overall ranking, possibly reflecting targeted training on academic literature. kimi-k2.5 (69.1%) demonstrates that open-weight models are approaching closed-source performance on domain-specific benchmarks. Critically, within-family performance generally correlates with model version (gpt-5.4 > gpt-5.2 > gpt-5 > gpt-5.1), validating that AER-bench effectively captures incremental capability improvements. Per-task 95% bootstrap confidence intervals for all 16 models are reported in Appendix A (Table 4).

To preempt concerns that the ranking above might be an artifact of sampling noise, we additionally conduct pairwise paired-bootstrap significance tests on the Overall score for all $\binom{16}{2}$ model pairs (Appendix B, Table 6): every cross-tier comparison rejects equality at $p \leq 0.003$, while a small number of within-cluster pairs—most notably gpt-5.4 vs. claude-4.6-opus—remain statistically tied, providing a conservative significance frame for every claim we make about model rankings.

Figure 3 illustrates the variation in task difficulty and discriminative power across the benchmark. Literature Review presents the lowest median score (under 50%) with tight clustering, indicating that all models struggle uniformly with citation network reconstruction and argument structure recovery. Conversely, Mathematical Modeling exhibits high scores (median near 90%) with limited spread, suggesting that symbolic manipulation capabilities have largely converged across modern model families. Tasks like Economic Knowledge and Model Generation demonstrate the widest score ranges, serving as the primary differentiators between average and frontier AI systems in economic research contexts. See Appendix G for detailed performance analysis by task dimension.

4.2 LLM-as-Judge Quality

To validate the multi-judge evaluation framework detailed in Section 3, we analyze the scoring consistency between two independent judge models: claude-4.6-sonnet and gpt-5.2. Figure 4 shows

Table 3: Overall benchmark scores (in %) with item-level 95% bootstrap confidence intervals on the Overall column. Per-task columns show point estimates; bold values mark the best score in each column. Models are grouped by provider and sorted by Overall within each group.

Provider	Model	Overall	EK	MM	PA	LR	MG	EA	CI
OpenAI	gpt-5.4	75.0 [74.0, 76.1]	73.9	92.8	76.2	56.0	85.8	81.6	58.9
	gpt-5.2	73.0 [72.1, 74.0]	70.8	91.5	76.4	50.3	83.6	78.2	60.5
	gpt-5	71.0 [69.9, 72.0]	74.2	91.8	73.5	47.7	84.2	78.5	46.9
	gpt-5.1	70.0 [69.0, 71.0]	67.5	89.7	75.6	48.1	82.8	75.0	51.4
	gpt-5-mini	69.0 [68.0, 70.0]	72.2	90.8	71.1	48.1	79.2	76.8	45.1
	gpt-4o	54.0 [53.1, 55.0]	25.6	78.4	72.1	39.5	45.7	64.6	52.3
Anthropic	claude-4.6-opus	74.0 [73.0, 74.9]	71.5	89.4	76.1	52.5	78.1	77.7	72.7
	claude-3.7-sonnet	73.7 [72.8, 74.6]	67.0	88.6	76.8	50.7	81.7	78.2	73.0
	claude-4.6-sonnet	73.4 [72.5, 74.3]	66.7	88.8	76.3	50.6	81.8	76.9	72.9
Google	gemini-3.1-pro	71.6 [70.6, 72.5]	76.3	89.3	74.3	47.0	70.0	72.0	72.0
	gemini-3-flash	65.8 [64.9, 66.8]	74.8	86.5	77.0	46.2	62.4	72.3	41.6
	gemini-2.5-flash	64.9 [63.9, 65.9]	69.4	88.0	76.1	41.4	68.0	67.1	44.1
Others	kimi-k2.5	69.1 [68.0, 70.1]	68.9	87.4	73.5	47.3	71.9	76.3	58.3
	glm-5.1	65.5 [64.5, 66.6]	57.9	85.4	75.8	49.0	69.5	74.8	46.3
	doubao-seed-1.6	64.8 [63.8, 65.8]	66.0	84.9	74.7	42.2	61.5	72.7	51.4
	qwen3-vl-32b-instruct	61.3 [60.3, 62.3]	60.9	82.0	73.5	39.1	59.4	66.2	48.0

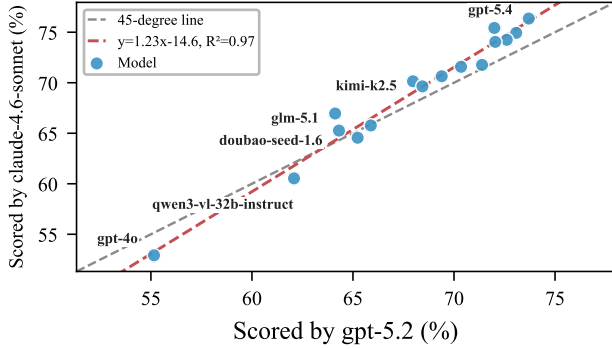


Figure 4: Judge agreement on overall benchmark scores across 16 models. Scores from two independent LLM judges (claude-4.6-sonnet and gpt-5.2) agree closely: the linear fit (dashed red, $y = 1.23x - 14.6$, $R^2 = 0.97$) follows the 45-degree line (dashed gray), and the mean absolute difference is only 1.6 percentage points, validating the robustness of the rubric-based evaluation framework.

high overall agreement, with a mean absolute difference of 1.6 percentage points. This tight clustering confirms that our dimension-level, criteria-bound rubrics effectively constrain the subjectivity typical of LLM evaluators, producing reliable assessments across different judge architectures.

While overall agreement is strong, Figure 5 reveals systematic task-level scoring preferences between the judges. claude-4.6-sonnet assigns higher scores on Mathematical Modeling (+4.0 pp) and

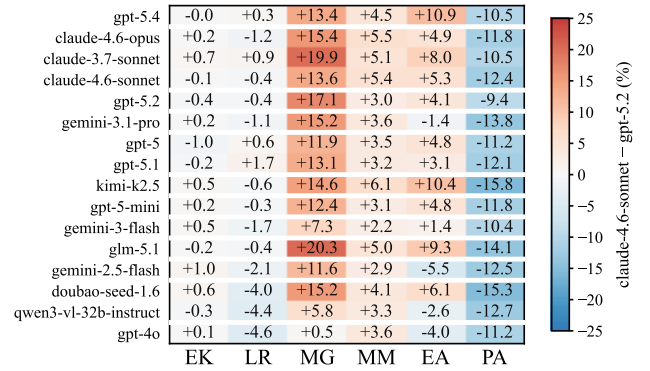


Figure 5: Task-level scoring differences of two judges. Heatmap shows the score difference (claude-4.6-sonnet minus gpt-5.2) across the six non-CI tasks, with red indicating higher scores from claude-4.6-sonnet and blue indicating higher scores from gpt-5.2. Models are sorted by overall performance, with provider boundaries marked by white lines.

Model Generation (+13.0 pp), whereas gpt-5.2 scores Paper Abstraction higher (-12.2 pp). To test whether these task-level offsets translate into a structural *self-preference bias*—each judge inflating scores for its own family—we computed the per-item Claude-minus-GPT score gap $\Delta_{m,i}$ for every examinee m , applied a paired Wilcoxon signed-rank test per model, and then compared the per-model mean Δ across families with a two-sided Mann-Whitney U test. Anthropic candidates do receive a larger “Claude-judge bonus” than OpenAI candidates (mean $\Delta = +2.90$ pp vs. +1.41 pp; the Google and “Other” families pooled together +0.82 pp), but the family-level difference falls short of conventional significance

($p = 0.095$, $n = 3$ vs. $n = 6$). The directional effect is real but small relative to the cross-tier gaps documented in Table 3, and arithmetic averaging across the two independent judges substantially attenuates it; per-model results and the full statistical analysis are reported in Appendix C.

4.3 Key Findings

Based on the quantitative results, we highlight three core findings. They should be read as module-level diagnostics of the model layer of a research agent: which stages a foundation model can carry on its own, and which require augmentation.

Finding 1: Knowledge-Dependent Modules Are the Bottleneck—Where Research Agents Need External Augmentation. Foundation models are volatile exactly in the stages of the workflow that depend on domain knowledge. Economic Knowledge (EK) scores span a 51-point range (25.6–76.3%), and Literature Review (LR) is uniformly weak—the strongest model reaches only 56.0%, with the lowest median across all tasks (Figure 3). Consistent with our design choice to isolate parametric knowledge (Section 2.4), these results identify precisely which stages of an agentic pipeline demand external augmentation: a research agent should not rely on its model core alone for domain facts or citation networks, but should back these stages with retrieval and citation databases, while reserving the model core for symbolic manipulation and synthesis, where performance is strong and stable.

Finding 2: Distinct Family-Level Capability Fingerprints. Beyond version-level scaling, AER-bench reveals task-specific signatures that distinguish model families. All five GPT-5 variants occupy the top five positions on MM (89.7–92.8%), with the leader gpt-5.4 ahead of the best non-GPT-5 model by 3.4 points; the 3 Claude models cluster tightly at 72.7–73.0% on CI—over 12 points above gpt-5.2—while staying within 0.6% overall, indicating a stable shared profile rather than checkpoint-level idiosyncrasy; and Gemini models trail GPT-5 and Claude by 3–23 points on LR, MG, and EA despite being competitive on EK and PA. These cross-family patterns are systematic rather than random, showing that AER-bench can diagnose the relative strengths of entire series, not merely rank isolated models.

Finding 3: An Uneven Grasp of Historical Economic Concepts. The Concept Identification (CI) task spans texts from 1900–1950, allowing us to probe models’ command of the history of economic thought. Aggregate scores run ~5 pp lower on 1900–1929 than on 1930–1950 chunks (Figure 6), led by the *equilibrium_scene* dimension. Robustness checks (Appendix D, Tables 8–9) sharpen the picture: after partialling out source-chunk length (OLS, model fixed effects, item-clustered standard errors), the $\text{Era}_2 - \text{Era}_1$ coefficient shrinks to +2.75 pp ($p = 0.44$); the gap lives almost entirely in the longest two length quartiles (Q3 +9.2, Q4 +12.7 pp), while the shortest two *reverse* (Era_1 leads by 1.8–2.1 pp). A per-decade $\times$ per-model breakdown then reveals strong heterogeneity: several frontier models score *higher* on 1900–1909 than on 1940–1950, whereas others show a canonical 20+ pp drop on the earliest decade. The honest characterization is therefore a *model-stratified, length-conditional* modernity bias: uneven rather than universal, but systematic enough to expose gaps in how current systems represent

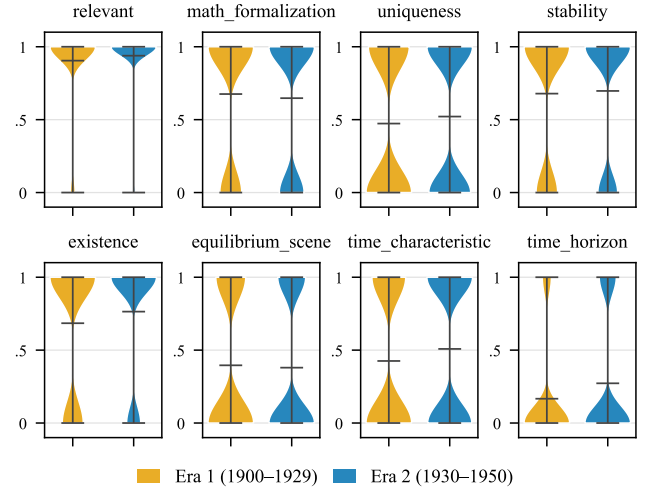


Figure 6: CI performance across historical periods. Violin plots show the distribution of 16 model scores on each of the 8 concept dimensions, comparing publications from 1900–1929 (orange) versus 1930–1950 (blue). Models exhibit systematic performance gaps on earlier texts, particularly for *equilibrium_scene* and *existence* dimensions.

the history of economic thought, likely reflecting pre-training corpus composition rather than uniform conceptual drift. The CI task serves as a diagnostic for which models still carry this blind spot.

5 Limitations

While AER-bench establishes a rigorous evaluation framework, it presents several inherent limitations. First, the automated task generation pipeline is computationally expensive. Processing full-length papers through multi-turn interactions requires top-tier models, currently costing approximately \$6 per paper. Consequently, we deliberately prioritized high-fidelity task extraction over massive scale. Second, the reliance on PDF parsing occasionally introduces formatting artifacts that can disrupt the extraction of complex mathematical notations. Third, although our dimension-level rubrics mitigate subjective biases, the LLM-as-Judge paradigm may still harbor residual architectural preferences. Fourth, the modern corpus consists predominantly of English-language journals, potentially masking disparities in models’ multilingual economic reasoning capabilities. Fifth, the benchmark does not include a human expert baseline, a decision motivated by several practical and conceptual challenges discussed in Appendix E. Finally, by design AER-bench evaluates the model layer rather than complete agentic systems (Section 2.4); quantifying the gains of retrieval and tool augmentation on top of the tool-free baselines reported here is an important direction for future work.

6 Conclusion

Can foundation models really understand economic research? This paper addresses the question through AER-bench, the first benchmark that evaluates foundation models across the full workflow

of economic research—the intellectual core of agentic economic research systems. Grounded in 191 top-journal articles and 900 expert-annotated pre-1950 segments, AER-bench assesses 16 frontier models on seven workflow stages, from domain knowledge and literature positioning to formal modeling, analysis, and written synthesis, using a hybrid framework that combines rule-based scoring with two independent LLM judges.

The answer it yields is sharply asymmetric: models have largely mastered the formal stages of research—mathematical modeling and written synthesis—but not the knowledge-dependent stages, where scholarly understanding largely resides. Domain knowledge and citation construction remain volatile, specifying precisely where a research agent requires external retrieval, while consistent family-level capability fingerprints and an uneven grasp of historical economic concepts expose gaps in models’ command of the history of economic thought. These findings highlight the necessity of moving from parametric model cores to augmented agentic systems, and AER-bench provides the evaluation layer for that transition: the tool-free baselines reported here are the reference points against which complete agentic research systems—such as the agentic frameworks that motivated this work—can be measured on the same tasks. Ultimately, AER-bench offers a systematic yardstick for the next generation of AI systems capable of conducting, interpreting, and advancing rigorous economic research.

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A Full Per-Task Ranking with Bootstrap Confidence Intervals

To complement the main-text ranking (Table 3), we report in Table 4 the full per-task results for all 16 evaluated models, with item-level 95% bootstrap confidence intervals on both the Overall score and every individual task. Confidence intervals are obtained from $B = 20,000$ bootstrap resamples stratified within each task; the same resample indices are shared across models so that pairwise comparisons preserve per-item correlation. For the five open-ended tasks (LR, MG, MM, EA, PA), the per-item score is the arithmetic mean of the `claude-4.6-sonnet` and `gpt-5.2` judges before resampling; EK and CI use rule-based scoring.

Table 4: Per-task and overall scores (in %) with item-level 95% bootstrap confidence intervals ($B = 20,000$ resamples, stratified within task). Each cell stacks the point estimate above its $[ci_{lo}, ci_{hi}]$ interval. Bold marks the column-wise maximum. Models are grouped by provider and sorted by Overall within each group.

Provider	Model	Overall	EK	MM	PA	LR	MG	EA	CI
OpenAI	gpt-5.4	75.0 [74.0, 76.1]	73.9 [72.4, 75.4]	92.8 [92.0, 93.5]	76.2 [73.9, 78.5]	56.0 [54.7, 57.3]	85.8 [83.2, 88.1]	81.6 [80.9, 82.3]	58.9 [53.0, 64.8]
	gpt-5.2	73.0 [72.1, 74.0]	70.8 [69.3, 72.2]	91.5 [90.7, 92.3]	76.4 [74.7, 78.0]	50.3 [49.2, 51.3]	83.6 [81.3, 85.8]	78.2 [77.7, 78.6]	60.5 [54.5, 66.5]
	gpt-5	71.0 [69.9, 72.0]	74.2 [72.5, 75.8]	91.8 [91.0, 92.6]	73.5 [71.2, 75.6]	47.7 [46.9, 48.6]	84.2 [81.5, 86.8]	78.5 [78.0, 78.9]	46.9 [40.9, 53.0]
	gpt-5.1	70.0 [69.0, 71.0]	67.5 [65.9, 69.0]	89.7 [88.7, 90.6]	75.6 [73.6, 77.5]	48.1 [47.1, 49.0]	82.8 [80.6, 84.8]	75.0 [74.4, 75.6]	51.4 [45.5, 57.2]
	gpt-5-mini	69.0 [68.0, 70.0]	72.2 [70.7, 73.7]	90.8 [89.8, 91.6]	71.1 [69.1, 73.1]	48.1 [47.0, 49.2]	79.2 [76.8, 81.5]	76.8 [76.2, 77.3]	45.1 [39.3, 51.0]
	gpt-4o	54.0 [53.1, 55.0]	25.6 [23.2, 28.2]	78.4 [77.0, 79.8]	72.1 [70.9, 73.2]	39.5 [38.6, 40.4]	45.7 [43.7, 47.7]	64.6 [63.9, 65.4]	52.3 [46.6, 58.0]
Anthropic	claude-4.6-opus	74.0 [73.0, 74.9]	71.5 [69.9, 73.0]	89.4 [88.4, 90.4]	76.1 [74.2, 77.9]	52.5 [51.4, 53.7]	78.1 [75.4, 80.7]	77.7 [77.1, 78.2]	72.7 [67.4, 77.8]
	claude-3.7-sonnet	73.7 [72.8, 74.6]	67.0 [65.4, 68.5]	88.6 [87.6, 89.7]	76.8 [75.1, 78.4]	50.7 [49.5, 51.8]	81.7 [79.5, 83.7]	78.2 [77.6, 78.7]	73.0 [67.6, 78.0]
	claude-4.6-sonnet	73.4 [72.5, 74.3]	66.7 [65.1, 68.3]	88.8 [87.7, 89.8]	76.3 [74.5, 78.0]	50.6 [49.4, 51.7]	81.8 [79.4, 84.1]	76.9 [76.3, 77.5]	72.9 [67.6, 78.0]
Google	gemini-3.1-pro	71.6 [70.6, 72.5]	76.3 [75.0, 77.6]	89.3 [88.3, 90.2]	74.3 [71.5, 77.0]	47.0 [46.2, 47.9]	70.0 [67.2, 72.6]	72.0 [71.5, 72.6]	72.0 [67.0, 76.8]
	gemini-3-flash	65.8 [64.9, 66.8]	74.8 [73.4, 76.2]	86.5 [85.5, 87.6]	77.0 [75.7, 78.1]	46.2 [45.2, 47.2]	62.4 [60.4, 64.4]	72.3 [71.7, 72.9]	41.6 [35.9, 47.4]
	gemini-2.5-flash	64.9 [63.9, 65.9]	69.4 [67.9, 71.0]	88.0 [87.0, 89.0]	76.1 [74.0, 78.0]	41.4 [40.5, 42.3]	68.0 [65.4, 70.5]	67.1 [66.5, 67.7]	44.1 [38.4, 50.0]
Others	kimi-k2.5	69.1 [68.0, 70.1]	68.9 [67.3, 70.5]	87.4 [86.2, 88.5]	73.5 [70.8, 75.9]	47.3 [46.1, 48.4]	71.9 [68.9, 74.7]	76.3 [75.6, 77.0]	58.3 [52.7, 63.9]
	glm-5.1	65.5 [64.5, 66.6]	57.9 [55.3, 60.5]	85.4 [84.1, 86.7]	75.8 [73.3, 78.2]	49.0 [47.8, 50.2]	69.5 [66.9, 72.0]	74.8 [74.0, 75.6]	46.3 [40.5, 52.0]
	doubao-seed-1.6	64.8 [63.8, 65.8]	66.0 [64.4, 67.6]	84.9 [83.7, 86.2]	74.7 [72.2, 77.1]	42.2 [41.2, 43.3]	61.5 [58.9, 64.2]	72.7 [72.0, 73.4]	51.4 [45.8, 57.0]
	qwen3-vl-32b-instruct	61.3 [60.3, 62.3]	60.9 [59.3, 62.5]	82.0 [80.6, 83.4]	73.5 [71.3, 75.5]	39.1 [38.3, 39.9]	59.4 [57.5, 61.3]	66.2 [65.5, 66.9]	48.0 [42.3, 53.8]

No single model dominates all dimensions. While `gpt-5.4` attains the column-wise maximum on five of the eight columns (Overall, MM, LR, MG, EA), it is decisively beaten on the remaining three: `gemini-3.1-pro` leads on EK (76.3 vs. 73.9), `gemini-3-flash` leads on PA (77.0 vs. 76.2), and the Anthropic family leads on CI by a wide margin (`claude-3.7-sonnet` 73.0; `claude-4.6-sonnet` 72.9; `claude-4.6-opus` 72.7) versus `gpt-5.4` at 58.9. The CI column in particular splits cleanly along provider lines: every Anthropic model exceeds 72%, whereas the OpenAI family except `gpt-4o` (52.3) sits between 45.1 and 60.5. Combined with the comparable ranking of `gemini-3.1-pro` (72.0) on the same task, this pattern suggests that performance on early-20th-century economic-history concepts depends more on pre-training corpus composition than on raw scale.

Confidence-interval width tracks item count and judge type. CI widths are far from uniform across tasks. The two largest tasks, EA ($n = 380$) and MM ($n = 399$), produce the tightest intervals: half-widths are typically 0.45–0.8 pp on EA and 0.75–1.4 pp on MM (e.g., `gpt-5.4` on EA has [80.9, 82.3]). The CI task is the opposite extreme, with half-widths near 5–6 pp on every model (e.g., `claude-4.6-opus` [67.4, 77.8]), reflecting both its smaller sample ($n = 160$) and the higher per-item variance of multi-field rule-based scoring. LR ($n = 187$) and MG ($n = 124$) show intermediate widths around 1–3 pp. These differences are quantitative reminders that gaps of 1–2 pp on CI should not be over-interpreted, whereas similar-sized gaps on EA or MM are typically well outside the bootstrap noise floor.

Anthropic models occupy a tight cluster. The three Claude models lie within 0.6 pp on Overall (74.0, 73.7, 73.4) and their per-task CIs overlap extensively on every column. Their largest within-family spread is on EK (66.7–71.5), where `claude-4.6-opus` clearly leads, and on MG (78.1–81.8), where the two Sonnet variants slightly exceed Opus—suggesting the Sonnet line carries a marginal advantage on procedural model construction. By contrast, the OpenAI line shows a near-monotonic version-vs-score relationship on Overall (`gpt-5.4` > `gpt-5.2` > `gpt-5` > `gpt-5.1` > `gpt-5-mini` > `gpt-4o`), with `gpt-4o` a clear outlier on EK at 25.6, more than 25 pp below every other model and consistent with its older training cutoff.

Where the rankings are statistically separable. Reading the bootstrap intervals horizontally provides an informal visual significance test on the Overall column: the two top-ranked models (`gpt-5.4` [74.0, 76.1] and `claude-4.6-opus` [73.0, 74.9]) overlap by roughly 1 pp, consistent with the paired-bootstrap p -value of 0.099 reported in the main text. The next three models (`claude-3.7-sonnet`, `claude-4.6-sonnet`,

gpt-5.2) form a tightly overlapping cluster between 72.1 and 74.6. After gemini-3.1-pro the intervals start to separate cleanly, with gpt-4o ([53.1, 55.0]) sitting more than 6 pp below the next-lowest model. Per-task intervals tell the same story at finer resolution: many cross-provider gaps on EK and CI are highly significant (no overlap at all), while LR and PA gaps are typically within the noise floor.

B Pairwise Paired-Bootstrap Significance on Overall Score

The point-estimate ranking in Table 3 is informative but does not by itself establish whether two adjacent models differ in a statistically meaningful sense. Table 6 reports the two-sided p -value for each ordered pair (i, j) of models under a paired bootstrap on the Overall score. For each of the $B = 20,000$ resamples we use the *same* per-task item indices to recompute every model’s Overall score, so that the difference $D_b = \widehat{O}_i^{(b)} - \widehat{O}_j^{(b)}$ inherits the per-item correlation between the two models. The reported p -value is the standard two-sided percentile, $\hat{p} = 2 \cdot \min(\Pr(D_b \geq 0), \Pr(D_b \leq 0))$, where the probabilities are taken over the bootstrap distribution. Bold cells flag $p < 0.05$; italic cells flag $0.05 \leq p < 0.10$; plain cells are non-significant. To keep the matrix readable, models are referenced by the integer IDs defined in Table 5; IDs follow the descending Overall ranking used throughout the paper.

Table 5: Model ID legend used in Table 6.

ID	Model	ID	Model
1	gpt-5.4	9	kimi-k2.5
2	claude-4.6-opus	10	gpt-5-mini
3	claude-3.7-sonnet	11	gemini-3-flash
4	claude-4.6-sonnet	12	glm-5.1
5	gpt-5.2	13	gemini-2.5-flash
6	gemini-3.1-pro	14	doubao-seed-1.6
7	gpt-5	15	qwen3-vl-32b-instruct
8	gpt-5.1	16	gpt-4o

The top of the ranking is a statistical tie. The most consequential pair for the headline claim is gpt-5.4 (ID 1) versus claude-4.6-opus (ID 2): their paired-bootstrap p -value is 0.099, which fails to reject equality at the 0.05 level and only barely crosses the 0.10 threshold. gpt-5.4 is significantly above the third-place claude-3.7-sonnet ($p_{1,3} = 0.023$) and claude-4.6-sonnet ($p_{1,4} = 0.011$), but its margin over claude-4.6-opus sits in the same regime as several within-cluster comparisons below. We therefore describe the leader as “statistically tied at the top” rather than as a clear winner in the main text.

Table 6: Pairwise paired-bootstrap two-sided p -values on the Overall benchmark score, $B = 20,000$ resamples with shared within-task item indices across models. All entries are reported in units of 10^{-3} , so a cell value of 92 means $p = 0.092$ and “ < 1 ” means $p < 10^{-3}$. Rows and columns are indexed by the model IDs listed in Table 5. Bold: $p < 0.05$. Italic: $0.05 \leq p < 0.10$. Plain: $p \geq 0.10$. The matrix is symmetric ($p_{ij} = p_{ji}$); the diagonal is empty.

[illegible]

A four-model cluster sits just below the leader. IDs 2–5 (claude-4.6-opus, claude-3.7-sonnet, claude-4.6-sonnet, gpt-5.2) are mutually indistinguishable: every pairwise comparison among them yields $p \in [0.153, 0.520]$, with the only borderline case being $p_{2,5} = 0.054$. The three Claude models in particular cannot be ordered statistically (all internal p -values ≥ 0.153). Combined with the leader pair above, this means the top five models (IDs 1–5) of Table 3 should be read as a *cluster* rather than as a strict ranking.

Lower-tier pairs that look close are also indistinguishable. Several mid-table neighbours fail to separate at conventional levels: $p_{6,7} = 0.343$ (gemini-3.1-pro vs. gpt-5), $p_{7,8} = 0.088$ (gpt-5 vs. gpt-5.1), $p_{9,10} = 0.980$ (kimi-k2.5 vs. gpt-5-mini), $p_{11,12} = 0.566$ (gemini-3-flash vs. glm-5.1), and $p_{13,14} = 0.834$ (gemini-2.5-flash vs. doubao-seed-1.6). These five clusters partition the leaderboard into five quasi-equivalence classes; differences within a class should not be treated as evidence of model superiority.

Cross-cluster gaps are robust. By contrast, every cross-cluster comparison—e.g. any leader-cluster model against any model below gemini-3.1-pro, or any mid-tier model against qwen3-vl-32b-instruct (ID 15) or gpt-4o (ID 16)—yields $p \leq 0.003$. The lowest two models are themselves strongly separated from each other ($p_{15,16} < 0.001$). Together these patterns indicate that AER-bench resolves coarse capability tiers unambiguously while leaving the finest within-tier ordering to be settled by additional data—a useful diagnostic for sample-size planning in future benchmark extensions.

C Judge Self-Preference Significance Analysis

The two LLM judges used for the open-ended generative tasks (claude-4.6-sonnet and gpt-5.2) are themselves candidates in the benchmark. Reviewers reasonably worry that this creates a structural conflict of interest: each judge could systematically inflate the scores of models from its own family. The main text (Section 4.2) summarises the result; this appendix documents the full procedure, the per-model statistics, and the limitations.

Statistical setup. For every examinee m and every item i on a non-CI task, we observe two normalised scores in $[0, 1]$, one from each judge. Define the per-item gap

$$\Delta_{m,i} = s_{m,i}^{\text{claude-judge}} - s_{m,i}^{\text{gpt-judge}},$$

so that $\Delta > 0$ means the Claude judge is more lenient on item i than the GPT judge. Stacking across the six non-CI tasks gives $n = 1,469$ paired observations per examinee. We run two complementary tests:

- (1) **Per-model paired Wilcoxon signed-rank test** on $\{\Delta_{m,i}\}_{i=1}^n$ (two-sided), to detect whether *any* systematic offset exists between the two judges for examinee m .
- (2) **Family-level Mann-Whitney U test** on the per-model means $\bar{\Delta}_m$, comparing the three Anthropic candidates against the six OpenAI candidates. This is the test that directly probes *in-family* bias: if the Claude judge favoured Anthropic candidates over OpenAI candidates, Anthropic models would have systematically larger $\bar{\Delta}_m$ than OpenAI models.

Per-model results. Table 7 reports $\bar{\Delta}_m$ and the Wilcoxon p -value for each of the 16 examinees. With $n = 1,469$ paired items the Wilcoxon test is highly powered, so almost every model shows a statistically significant non-zero gap; the magnitude rather than the p -value is the quantity of interest.

A directional but underpowered in-family effect. The three Anthropic candidates receive an average Claude-judge bonus of $\bar{\Delta} = +2.90$ pp, roughly twice the OpenAI candidates’ $+1.41$ pp and more than three times the “Other” families’ $+0.82$ pp. The direction of the effect is exactly what the self-preference hypothesis predicts. However, the Mann-Whitney U test on the per-model means yields $p = 0.095$, falling short of the conventional 0.05 threshold: with only $n_g = 3$ Anthropic and $n_g = 6$ OpenAI candidates the family-level test has limited power. We therefore *cannot* reject the null of no in-family bias, but we equally *cannot* claim that the apparent ~ 1.5 pp Anthropic–OpenAI gap is zero.

Why the effect is small in practice. Two facts bound the practical impact of this directional bias on the benchmark rankings. First, the quantity that matters for the leaderboard is the *between-family* component (~ 1.5 pp), not the global Claude-vs-GPT lenience offset ($+1.43$ pp averaged across all 16 candidates), because the latter is absorbed by averaging the two judges rather than picking either one. Second, the bias is far smaller than the leaderboard’s gross dynamic range: the top-to-bottom Overall gap is roughly 21 pp (gpt-5.4 75.0% vs. gpt-4o 54.0%), and the only Anthropic–OpenAI comparisons whose ordering a ~ 1.5 pp shift could plausibly disturb already sit inside the within-cluster ties that the pairwise paired-bootstrap analysis (Appendix B) flags as statistically indistinguishable (most prominently gpt-5.4 vs. claude-4.6-opus, $p = 0.099$). All Anthropic-vs-OpenAI pairs that *are* significant in Appendix B have point-estimate gaps that substantially exceed the residual family-level bias, so no cross-tier ranking claim in the main text is at risk of flipping under a 1.5 pp adjustment.

What this analysis does not establish. The two-judge protocol attenuates but does not eliminate the conflict-of-interest concern: the test above is necessary but not sufficient. A truly independent third-party judge—e.g. gemini-3.1-pro, which is a candidate but not currently a judge—would be needed to fully decouple judging from candidate selection. We treat the addition of a third independent judge as the single highest-value follow-up experiment for future versions of AER-bench, and discuss it explicitly in the limitations section.

Table 7: Per-examinee mean Claude-minus-GPT judge gap $\bar{\Delta}_m$ (in percentage points) with paired Wilcoxon signed-rank p -values (two-sided, $n = 1,469$ items per row). Positive values mean the Claude judge scores examinee m higher than the GPT judge does. The footnote reports family-level mean gaps and the Mann–Whitney U test comparing Anthropic vs. OpenAI candidates.

Examinee	Family	n	$\bar{\Delta}$ (pp)	Wilcoxon p
claude-3.7-sonnet	Anthropic	1469	+3.96	<0.001
claude-4.6-opus	Anthropic	1469	+2.41	<0.001
claude-4.6-sonnet	Anthropic	1469	+2.32	<0.001
gemini-2.5-flash	Google	1469	-1.43	<0.001
gemini-3-flash	Google	1469	+0.05	0.486
gemini-3.1-pro	Google	1469	-0.01	<0.001
gpt-4o	OpenAI	1469	-2.05	<0.001
gpt-5	OpenAI	1469	+1.69	<0.001
gpt-5-mini	OpenAI	1469	+1.60	<0.001
gpt-5.1	OpenAI	1469	+1.39	<0.001
gpt-5.2	OpenAI	1469	+2.00	<0.001
gpt-5.4	OpenAI	1469	+3.85	<0.001
doubao-seed-1.6	Other	1469	+1.56	<0.001
glm-5.1	Other	1469	+3.57	<0.001
kimi-k2.5	Other	1469	+3.53	<0.001
qwen3-vl-32b-instruct	Other	1469	-1.52	<0.001

Group means: Anthropic $\bar{\Delta} = +2.90$ pp ($n_g = 3$); OpenAI $\bar{\Delta} = +1.41$ pp ($n_g = 6$); Google+Other $\bar{\Delta} = +0.82$ pp ($n_g = 7$). Mann–Whitney p (Anthropic vs OpenAI) = 0.095.

D Robustness Controls for the Historical Concept-Drift Finding

The main-text Finding 3 (Section 4) reframes our Concept Identification (CI) results as a *model-stratified, length-conditional* “modernity bias” rather than a uniform property of contemporary foundation models. This appendix documents the two robustness controls behind the reframing: an item-clustered OLS regression with chunk-length control, and a per-decade $\times$ per-model breakdown of CI accuracy with bootstrap confidence intervals.

Why these controls are needed. The aggregate Era₂ – Era₁ gap of $\sim +5$ pp in Figure 6 can be confounded by at least three non-conceptual factors: (i) systematic length differences between eras; (ii) a 30-year era partition that may hide decade-level heterogeneity; and (iii) aggregation across 16 models with potentially very different behaviors. We re-analyze the CI data along all three axes.

Data and setup. We recover each CI item’s publication year and source-chunk text from the benchmark’s raw CI files, use chunk character count as a length proxy, and bin items into five decades. Of 160 CI items, 146 carry a usable year + chunk pair; the decade breakdown is {10, 22, 27, 43, 44} items. Per-model bootstrap intervals on each decade use $B = 10,000$ item-level resamples.

OLS with chunk-length control. We estimate the panel regression

$$\text{score}_{m,i} = \alpha_m + \beta_{\text{Era}_2} \mathbb{I}[\text{Era}_2]_i + \beta_{\log \text{len}} \log(\text{len})_i + \beta_{\times} \mathbb{I}[\text{Era}_2]_i \cdot \log(\text{len})_i + \varepsilon_{m,i},$$

with model fixed effects α_m , mean-centred $\log(\text{len})_i$, and item-clustered standard errors. We obtain $\hat{\beta}_{\text{Era}_2} = +0.0275$ ($p = 0.442$), $\hat{\beta}_{\log \text{len}} = -0.1502$ ($p < 10^{-5}$), $\hat{\beta}_{\times} = +0.0572$ ($p = 0.319$). Two facts follow: (a) the unconditional +5 pp era gap shrinks to +2.75 pp and is no longer distinguishable from zero once length is controlled for; (b) longer paragraphs are uniformly harder (~ -15 pp per unit of log-length), the dominant CI difficulty signal. The Era₂ $\times$ length interaction is positive but not individually significant.

Length-quartile stratification. Table 8 reveals a non-linear pattern the OLS coefficients hide: the per-quartile Era₂ – Era₁ gap is -1.8 pp on Q1, -2.1 pp on Q2, $+9.2$ pp on Q3, and $+12.7$ pp on Q4. The two shortest quartiles *reverse* the sign of the modernity bias, while the two longest account for essentially all of the aggregate +5 pp gap—consistent with representational gaps that surface only when models must reason over several inter-related sentences of pre-1930 prose.

Per-decade, per-model heterogeneity. Table 9 shows that aggregating across models obscures the dominant regularity. Along the 1900–1909 vs. 1940–1950 axis, models split into two clean groups:

- **Earliest-decade-strong** (claude-4.6-sonnet, claude-4.6-opus, claude-3.7-sonnet, gemini-3.1-pro, gpt-5.2, kimi-k2.5, gpt-5.1, doubao-seed-1.6, and qwen3-vl-32b-instruct) score *higher* on 1900–1909 than on 1940–1950, with old-to-recent drops between -1 and -22 pp (extreme: claude-4.6-sonnet 86.2% vs. 64.2%).
- **Modern-skewed** (gpt-5.4, gpt-5, gpt-5-mini, gemini-3-flash, gemini-2.5-flash, glm-5.1) show the canonical pattern: 19–41% on 1900–1909 rising roughly monotonically to 46–59% on 1940–1950; gpt-4o is essentially flat across decades (+2.6 pp).

Table 8: Length-stratified CI accuracy by era. “Gap” is the per-quartile Era₂ – Era₁ difference in mean accuracy; the bottom row reports the OLS coefficients. After controlling for length, the Era₂ main effect is no longer distinguishable from zero.

Length quartile	Era 1 acc. (%)	Era 2 acc. (%)	Gap (pp)	n_{Era1}	n_{Era2}
Q1 (shortest)	69.1	67.3	-1.8	304	288
Q2	49.4	47.3	-2.1	240	336
Q3	48.3	57.5	+9.2	256	320
Q4 (longest)	42.3	55.0	+12.7	144	448
OLS with model fixed effects and item-clustered standard errors:					
$\beta_{Era2} = +0.0275$ ($p = 0.442$);					
$\beta_{\log len} = -0.1502$ ($p = 9.16 \times 10^{-6}$);					
$\beta_{Era2 \times \log len} = +0.0572$ ($p = 0.319$).					

Table 9: Per-decade CI accuracy (%) with item-level 95% bootstrap confidence intervals ($B = 10,000$). Models are ordered by their 1900–1909 score; the *Items* row gives the per-decade sample size, so intervals on the $n = 10$ column are correspondingly wide.

Model	1900-09	1910-19	1920-29	1930-39	1940-50
<i>Items</i>	10	22	27	43	44
claude-4.6-sonnet	86.2 [65.0, 100.0]	80.1 [66.5, 92.6]	78.2 [66.7, 88.4]	67.2 [57.6, 76.2]	64.2 [52.8, 75.6]
claude-4.6-opus	82.5 [62.5, 97.5]	83.5 [71.0, 94.3]	69.4 [54.6, 82.9]	69.5 [59.9, 78.5]	65.3 [54.0, 75.9]
gemini-3.1-pro	81.2 [58.8, 96.2]	75.6 [59.1, 90.3]	75.0 [63.4, 86.1]	63.7 [54.1, 72.7]	72.2 [63.6, 80.1]
claude-3.7-sonnet	76.2 [48.8, 97.5]	80.7 [66.5, 92.6]	79.2 [67.1, 89.4]	64.8 [54.4, 75.0]	67.3 [55.7, 77.6]
doubao-seed-1.6	66.2 [41.2, 87.5]	45.5 [27.8, 63.6]	54.2 [39.8, 68.5]	45.6 [36.3, 54.9]	51.7 [41.8, 61.9]
gpt-5.1	65.0 [40.0, 86.2]	51.1 [33.0, 69.3]	42.6 [28.2, 57.4]	52.3 [42.7, 61.9]	49.7 [38.9, 60.5]
gpt-5.2	61.3 [32.5, 86.2]	66.5 [47.7, 83.5]	68.1 [53.2, 81.9]	57.0 [46.8, 67.2]	54.8 [44.0, 65.9]
kimi-k2.5	61.3 [33.8, 85.0]	51.7 [33.5, 69.9]	61.1 [46.8, 74.5]	58.4 [49.1, 67.4]	57.4 [47.4, 67.3]
gpt-4o	50.0 [22.5, 76.2]	44.9 [27.3, 63.6]	51.9 [37.0, 66.7]	57.3 [47.4, 66.6]	52.6 [41.8, 62.8]
qwen3-vl-32b-instruct	46.2 [20.0, 72.5]	42.6 [25.0, 60.8]	51.9 [37.0, 66.2]	50.0 [40.4, 59.6]	45.2 [35.2, 55.7]
gpt-5.4	41.2 [15.0, 68.8]	52.8 [35.2, 71.0]	56.9 [41.7, 71.8]	64.8 [55.8, 73.8]	59.4 [48.0, 70.5]
gpt-5	31.2 [7.5, 57.5]	31.8 [15.9, 49.4]	45.8 [31.5, 60.6]	52.9 [42.1, 63.1]	56.2 [45.2, 67.6]
glm-5.1	31.2 [6.2, 57.5]	31.8 [15.9, 48.9]	45.4 [31.0, 59.7]	48.0 [37.8, 57.8]	55.7 [45.7, 65.9]
gemini-2.5-flash	30.0 [6.2, 56.2]	29.5 [14.2, 46.6]	43.1 [28.7, 57.4]	48.8 [38.1, 58.7]	52.8 [42.3, 63.4]
gpt-5-mini	18.8 [0.0, 38.8]	29.0 [14.2, 45.5]	47.2 [32.9, 62.0]	53.2 [43.3, 63.1]	46.9 [35.8, 57.7]
gemini-3-flash	18.8 [0.0, 37.5]	32.4 [15.9, 50.0]	40.3 [25.9, 54.2]	49.4 [38.7, 59.9]	46.3 [34.9, 57.1]

Earliest-decade intervals are wide ($n = 10$), so the ~ 20 pp within-row swings should be read alongside the 1–2 pp swings on the densely-sampled 1940–1950 column.

What the reframing buys us. Together, the three controls support a much narrower claim than “current AI systems exhibit historical concept drift”: the era main effect vanishes after length control ($p = 0.442$); the descriptive gap concentrates in the longest two quartiles; and the per-decade structure differs sharply by model, with frontier systems such as claude-4.6-sonnet, gemini-3.1-pro, and gpt-5.2 scoring *better* on the earliest decade than on the most recent. We therefore frame the CI task as a *diagnostic for which models still carry a representational blind spot on early-20th-century economic prose*, rather than as evidence of a uniform property of contemporary foundation

models. The same controls flag the two most informative follow-ups: (i) inter-annotator agreement on the 1900–1909 subset, where bootstrap intervals are widest, and (ii) controlled experiments holding chunk length fixed across eras.

E Why AER-bench Omits a Human Expert Baseline

A natural question when presenting a benchmark like AER-bench is how foundation models compare to human experts. After careful consideration, we have chosen not to collect a human performance baseline. This appendix details the reasons behind that decision.

Economic expertise is deeply fragmented. Modern economics is an enormously broad discipline, spanning macroeconomics, microeconomic theory, econometrics, labor, development, industrial organization, international trade, finance, and many other fields. A typical economics PhD student or even an experienced researcher usually possesses deep knowledge in only one or two of these areas. AER-bench draws tasks from 191 top-5 journal articles covering the full spectrum of contemporary economics. Consequently, no single human expert can be expected to perform competently across all seven task types and all subfields. Assembling a panel of specialists who could collectively cover the entire benchmark would not only be logistically daunting but would also introduce severe inter-annotator variability, making a single “human baseline” number difficult to interpret.

Human experts routinely rely on tools. In real-world research, economists seldom work from memory alone. They consult literature databases, use statistical software, and increasingly interact with large language models themselves. If we were to allow human participants to use such tools in order to simulate realistic research conditions, the resulting “human” baseline would in fact be a hybrid human–AI system. That would largely defeat the purpose: our aim is to measure how far pure foundation models are from performing end-to-end economic research, not to compare them against a human–AI combination that is not under controlled conditions.

Restricting human experts to a closed-book setting is unrealistic and difficult to enforce. Conversely, if we were to forbid any external aids, we would be testing an artificial skill—unaided recall of paper details—that is not how economists actually work. Moreover, ensuring that remotely participating experts do not quickly search for answers is practically impossible. The resulting data would be noisy and likely unrepresentative of genuine human expertise.

The benchmark already embeds a form of expert knowledge. It is worth noting that AER-bench already incorporates human expertise in two critical ways. First, all Concept Identification tasks are manually annotated by economics doctoral students, providing gold-standard labels. Second, the LLM-as-Judge rubrics are designed and weighted to reflect what a domain expert would value in an answer. These design choices ensure that the evaluation criteria are grounded in expert judgment, even without a separate human performance baseline.

Conclusion. Given these considerations—the breadth of expertise required, the confounding effect of tool use, the impracticality of pure closed-book testing, and the expert knowledge already embedded in the benchmark design—we believe that adding a human baseline would not strengthen the actionable insights that AER-bench provides. We instead focus on comparisons across models, which already reveal consistent performance hierarchies and systematic weaknesses. We encourage future work to explore targeted human–model comparisons on narrower subsets of the benchmark where domain specialists can be meaningfully recruited.

F Detailed Rubric Definitions

This appendix provides detailed rubric definitions for all LLM-as-Judge tasks in AER-bench. Each task employs multi-dimensional scoring to prevent holistic biases and ensure fine-grained evaluation of model capabilities.

F.1 Literature Review (LR)

Literature Review evaluates the ability to reconstruct an introduction section that aligns with the paper’s citation network and argumentation structure. This task employs a hybrid scoring approach combining citation recall and writing quality.

Citation Score (Weight: 1). Measures weighted recall of citations across three tiers:

- **Tier 1 (weight 0.8):** Core citations indispensable to the argument, typically from top journals or highly influential sources.
- **Tier 2 (weight 0.2):** Supplementary citations from reputable academic sources.
- **Tier 3 (weight 0.0):** Optional citations with lower relevance or authority; not scored.

The citation score is computed as: $\text{citation_score} = 0.8 \times \text{Tier1_recall} + 0.2 \times \text{Tier2_recall}$.

Rubric Score (Weight: 1). Assesses writing quality across paper-specific dimensions tailored to the introduction’s argumentation structure. Common dimensions include:

- **Research question clarity:** Is the core question clearly stated?
- **Literature positioning:** Does the introduction position the paper within existing literature?
- **Contribution statement:** Are the paper’s contributions explicitly articulated?
- **Methodological preview:** Is the research approach previewed appropriately?
- **Logical flow:** Does the introduction follow a coherent argumentative structure?

Each dimension is scored 0–10, with weights summing to 100. The rubric is dynamically constructed for each paper based on its specific introduction structure.

F.2 Model Generation (MG)

Model Generation evaluates the ability to construct a complete formal economic model from a desymbolized narrative description. Unlike other tasks, MG uses a flexible element-based rubric where the number and type of required elements vary by question.

Element Rubric (Flexible Dimensions). Each model is decomposed into constituent elements (agents, objective functions, constraints, equilibrium concepts, first-order conditions). Each element is scored based on presence and correctness:

- **Full (100%):** Element is present, correctly formalized, uses proper mathematical notation, and is economically meaningful.
- **Substantial (70%):** Element is present and mostly correct, but has minor issues (e.g., missing a variable, slight notation inconsistency, or incomplete specification).
- **Partial (40%):** Element is mentioned or attempted, but has significant issues (e.g., wrong functional form, missing key variables, or conceptually flawed formalization).
- **Minimal (15%):** Element is only vaguely referenced without proper formalization.
- **Missing (0%):** Element is not present at all.

The total score is the sum of all element scores, normalized to 0–100. The maximum possible score and element weights are determined by the reference model’s complexity. For example, a model with 2 agents, 2 objective functions, 3 constraints, 1 equilibrium concept, and 2 FOCs would have 10 elements, each contributing proportionally to the total score.

F.3 Mathematical Modeling (MM)

Mathematical Modeling evaluates the ability to derive equations, solve optimization problems, and perform formal mathematical reasoning in economic contexts. Each question is scored across four dimensions (0–10 points each, total 40 points):

Correctness (Weight: 4). Assesses whether formulas are mathematically equivalent to the reference answer and derivation steps are valid.

- **9–10:** All formulas are mathematically equivalent to the reference; derivation steps are valid.
- **6–8:** Minor errors (sign, index, simplification) but core structure is correct.
- **3–5:** Significant errors but partial understanding of the approach is evident.
- **0–2:** Fundamentally incorrect or mostly missing.

Notation Consistency (Weight: 2). Evaluates whether symbols match the problem setup and are used consistently throughout the derivation.

- **9–10:** All symbols match the problem setup and are used consistently.
- **6–8:** Minor inconsistencies that do not affect readability.
- **3–5:** Some symbols undefined or used inconsistently.
- **0–2:** Major notation confusion or deviation from given symbols.

Completeness (Weight: 3). Checks whether all intermediate steps are shown and a reader can verify each step.

- **9–10:** All intermediate steps shown; a reader can verify each step.
- **6–8:** Key steps present but some intermediate algebra omitted.
- **3–5:** Major steps missing; jumps to conclusions without justification.
- **0–2:** Only final answer given, or derivation is fragmentary.

Economic Intuition (Weight: 1). Assesses whether the candidate provides clear interpretation of results in economic terms and explains what the mathematics means.

- **9–10:** Clear interpretation of results in economic terms; explains what the math means.
- **6–8:** Some economic interpretation provided.
- **3–5:** Minimal economic context; mostly mechanical derivation.
- **0–2:** No economic interpretation at all.

F.4 Economic Analysis (EA)

Economic Analysis evaluates the ability to interpret empirical or theoretical results with substantive economic reasoning. Given a purely factual data description (e.g., regression coefficients, significance levels, trends), the model must produce a professional analysis paragraph. Scoring is based on six dimensions (0–10 points each), with applicability determined per question:

Accuracy (Weight: 2). Assesses whether numerical descriptions and trend characterizations are factually correct.

- **9–10:** All numerical and directional claims are accurate; no fabrication.
- **6–8:** Minor inaccuracies that do not fundamentally mislead.
- **3–5:** Some factual errors in data description.

- **0–2:** Major errors or fabricated numbers.

Economic Interpretation (Weight: 2). Evaluates whether the analysis provides substantive economic meaning (e.g., magnitude interpretation, heterogeneity discussion).

- **9–10:** Rich interpretation of economic significance and practical implications.
- **6–8:** Reasonable interpretation provided.
- **3–5:** Superficial or generic interpretation.
- **0–2:** No substantive interpretation.

Mechanism (Weight: 2). Assesses whether economic channels or causal pathways are identified.

- **9–10:** Clear identification of driving mechanisms.
- **6–8:** Some mechanisms discussed but not fully developed.
- **3–5:** Vague or incomplete mechanism discussion.
- **0–2:** No mechanism identified.

Theory Connection (Weight: 1.5). Evaluates whether results are connected to theoretical predictions or prior work.

- **9–10:** Strong theoretical grounding with specific references.
- **6–8:** Some theoretical context provided.
- **3–5:** Weak or generic theory connection.
- **0–2:** No theory connection.

Policy Implication (Weight: 1.5). Assesses whether policy implications are discussed appropriately (only applicable when results support policy discussion).

- **9–10:** Nuanced, well-grounded policy discussion.
- **6–8:** Reasonable policy points raised.
- **3–5:** Superficial policy discussion.
- **0–2:** No policy discussion.
- **N/A:** Not applicable for this result.

Writing Quality (Weight: 1). Judges whether the analysis is well-structured and clearly written.

- **9–10:** Publication-ready prose; logical flow; precise language.
- **6–8:** Clear and organized; minor style issues.
- **3–5:** Readable but poorly organized or imprecise.
- **0–2:** Disorganized or unclear.

Note: Not all dimensions are applicable to every question. For example, a purely theoretical result may not have policy implications. The normalized score is computed as: $\text{normalized_score} = \frac{\text{total}}{\text{max_possible}}$, where max_possible counts only applicable dimensions (each worth 10 points).

F.5 Paper Abstraction (PA)

Paper Abstraction evaluates the ability to synthesize a condensed paper body (approximately 30% of original length, excluding the abstract) into a coherent abstract that captures key contributions, methods, and findings. Scoring is based on four dimensions (0–10 points each, total 40 points):

Coverage (Weight: 4). Measures whether all or nearly all reference key points (atomic semantic propositions extracted from the original abstract) are mentioned or clearly implied.

- **9–10:** All or nearly all reference key points are mentioned or clearly implied.
- **6–8:** Most key points covered; 1–2 minor omissions.
- **3–5:** Several important points missing.
- **0–2:** Only 1–2 points covered, or major elements absent.

Accuracy (Weight: 4). Assesses factual correctness of all stated claims per the paper content.

- **9–10:** All stated claims are factually correct per the paper content.
- **6–8:** Minor inaccuracies that do not fundamentally mislead.
- **3–5:** Some factual errors or misrepresentations of results.
- **0–2:** Major factual errors that distort the paper’s findings.

Conciseness (Weight: 3). Evaluates whether the abstract stays within the specified word limit with no redundancy.

- **9–10:** Within word limit; no redundancy; every sentence adds value.
- **6–8:** Mostly concise; minor redundancy or slightly over limit.
- **3–5:** Noticeably wordy or exceeds word limit significantly.
- **0–2:** Excessively long or highly redundant.

Academic Style (Weight: 1). Judges whether the abstract reads like a published abstract in a top economics journal.

- **9–10:** Reads like a published abstract in a top economics journal.
- **6–8:** Generally academic but with some informal phrasing.
- **3–5:** Mixed register; some casual language or poor structure.
- **0–2:** Not academic in style.

F.6 Judge Model Configuration

All LLM-as-Judge evaluations in AER-bench use the following judge models:

- **claude-4.6-sonnet** (Anthropic)
- **gpt-5.2** (OpenAI)

For each generative task, both judges independently score all dimensions. The final item score is the arithmetic mean of the two judges' scores. Section 4.2 presents a systematic comparison of inter-judge agreement, demonstrating a mean absolute difference of 1.6 percentage points across all models and tasks, validating the robustness of this multi-judge approach.

G Detailed Performance Analysis by Task Dimension

This appendix provides a comprehensive breakdown of model performance across all seven task dimensions in AER-bench. Figure 7 presents a visual summary of the overall ranking table (Table 3), showing task-level performance for all evaluated models grouped by provider. Each subsection then presents a heatmap visualization showing dimension-level scores for all evaluated models, followed by detailed analysis of performance patterns, strengths, and weaknesses.

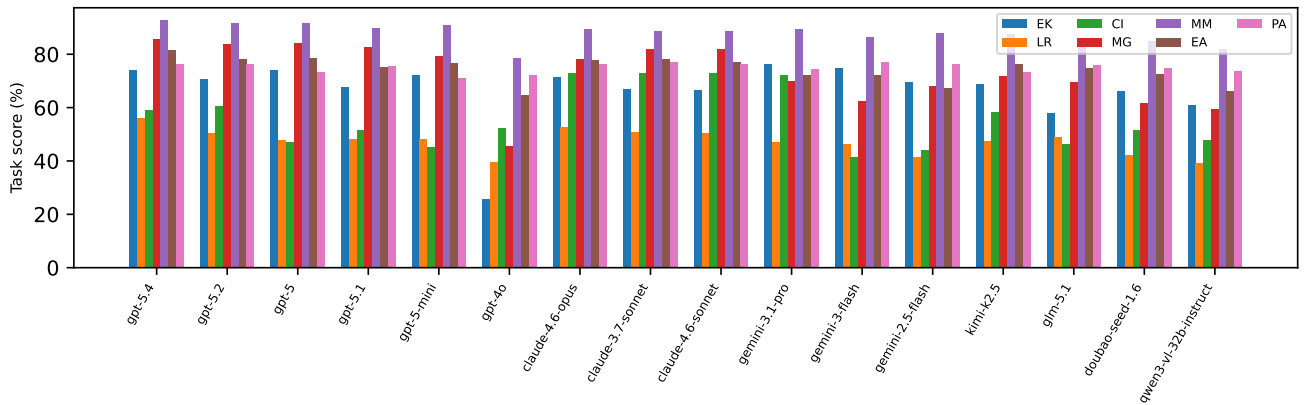


Figure 7: Task-level performance across all evaluated models, grouped by provider. This figure provides a visual complement to Table 3, showing the distribution of scores across the seven task dimensions (EK, LR, CI, MG, MM, EA, PA) for each model.

G.1 Economic Knowledge (EK)

The Economic Knowledge task evaluates models across four question formats: single choice, true/false, fill-in-blank, and ordering. As shown in Figure 8, performance varies significantly across these formats. Google's gemini-3.1-pro achieves the highest overall EK score (76.3%), demonstrating particularly strong performance on single-choice questions. OpenAI's GPT-5 series models show consistent performance across all four formats, with gpt-5.4 and gpt-5 both scoring above 73%.

Notably, fill-in-blank questions prove most challenging across all models, with average scores approximately 40–50 percentage points lower than single-choice questions (e.g., gemini-3.1-pro scores 92% on single choice but only 44% on fill-in-blank). This suggests that open-ended knowledge recall is fundamentally more difficult than recognition-based tasks. Single-choice questions generally yield the highest scores across models (up to 92%), followed by true/false (up to 90%) and ordering items (up to 80%).

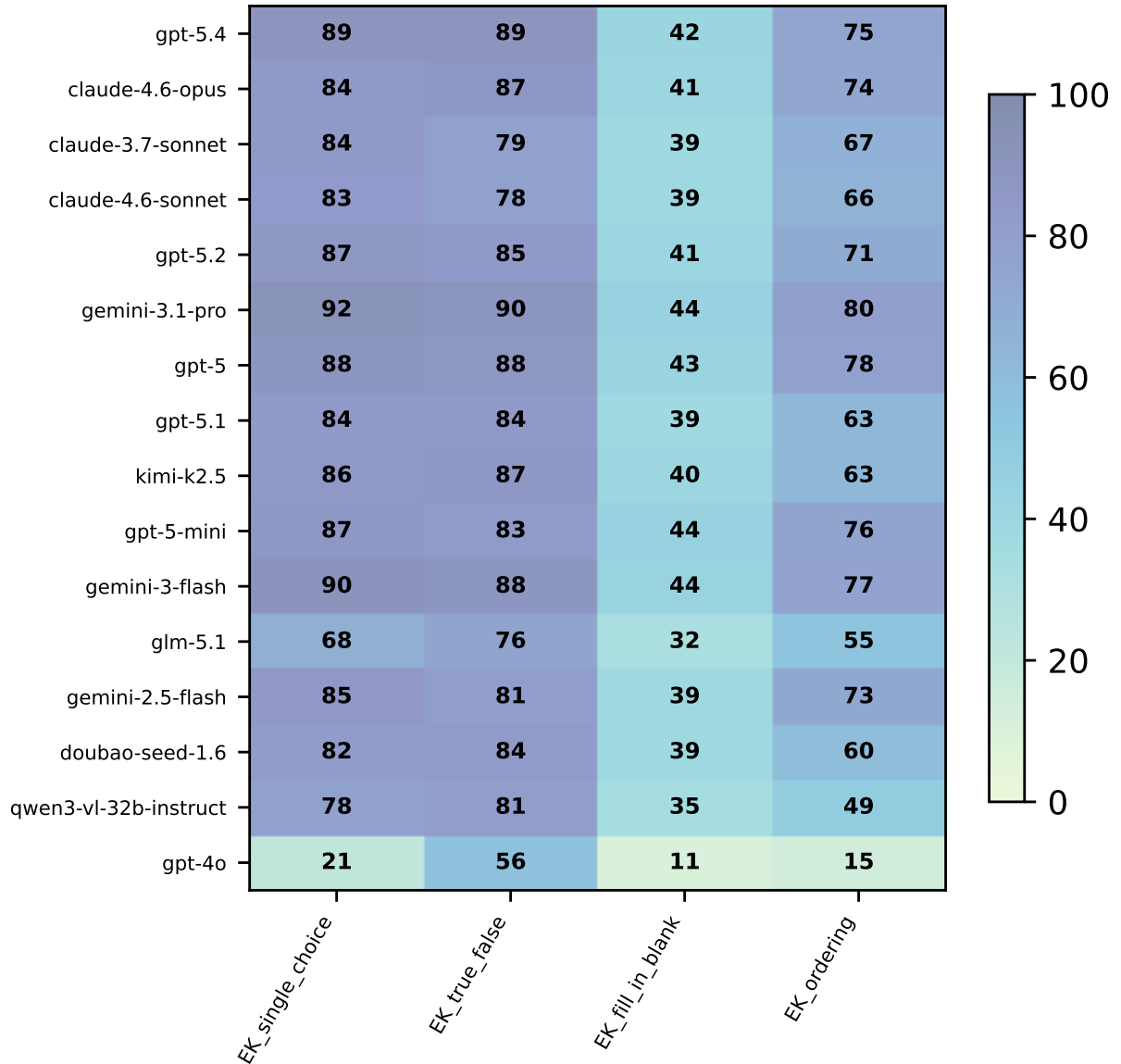


Figure 8: Performance breakdown across Economic Knowledge (EK) question types: Single Choice (SC), True/False (TF), Fill-in-Blank (FB), and Ordering (OD). Values represent percentage scores (0–100).

Among the lower-performing models, gpt-4o shows a marked weakness in this dimension (25.6% overall), particularly struggling with fill-in-blank and ordering tasks. This indicates that older model generations lack the economic domain knowledge depth required for AER-bench tasks.

G.2 Mathematical Modeling (MM)

Mathematical Modeling represents the highest-scoring task across all models, with most frontier models exceeding 85% overall performance. Figure 9 reveals that gpt-5.4 achieves exceptional performance (92.8%), with near-perfect scores across all four evaluation dimensions: correctness, notation, completeness, and economic intuition.

The heatmap shows remarkably consistent performance across dimensions for top-tier models. Correctness scores are universally high (>90%) for GPT-5 series models (gpt-5.4, gpt-5.2, gpt-5 all at 92.2%, gpt-5-mini at 91.3%, gpt-5.1 at 90.3%), indicating strong

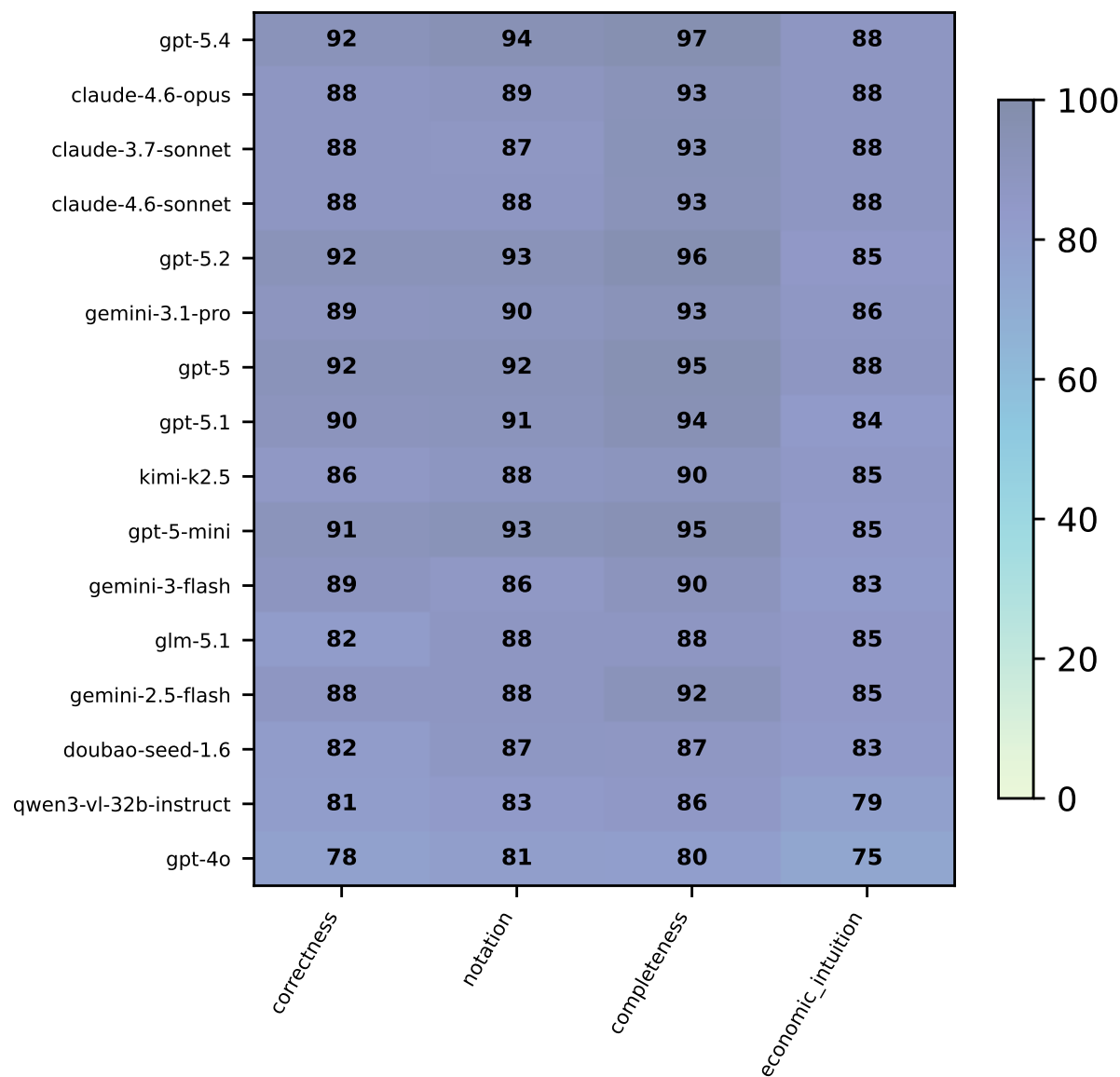


Figure 9: Performance breakdown across Mathematical Modeling (MM) dimensions: Correctness (Corr), Notation (Notn), Completeness (Comp), and Economic Intuition (Intu). Values represent percentage scores (0–100).

mathematical reasoning capabilities. Claude 4.6 models achieve slightly lower but still strong correctness scores (87.7–87.9%). Notation scores, which assess proper use of mathematical symbols and formatting, are similarly high, suggesting models have learned standard economic notation conventions from training data.

Completeness and economic intuition dimensions show slightly more variation. Economic intuition, which evaluates whether models provide meaningful economic interpretation alongside mathematical derivations, proves most challenging. Models like `gemini-3-flash` (6.1 points) and `gpt-5-mini` (6.6 points) show a 5–8 point gap between correctness and intuition scores, indicating they can solve equations correctly but struggle to explain economic meaning.

The uniformly high MM scores across models suggest that mathematical manipulation and symbolic reasoning are well-developed capabilities in current foundation models, likely benefiting from extensive training on mathematical and scientific texts.

G.3 Paper Abstraction (PA)

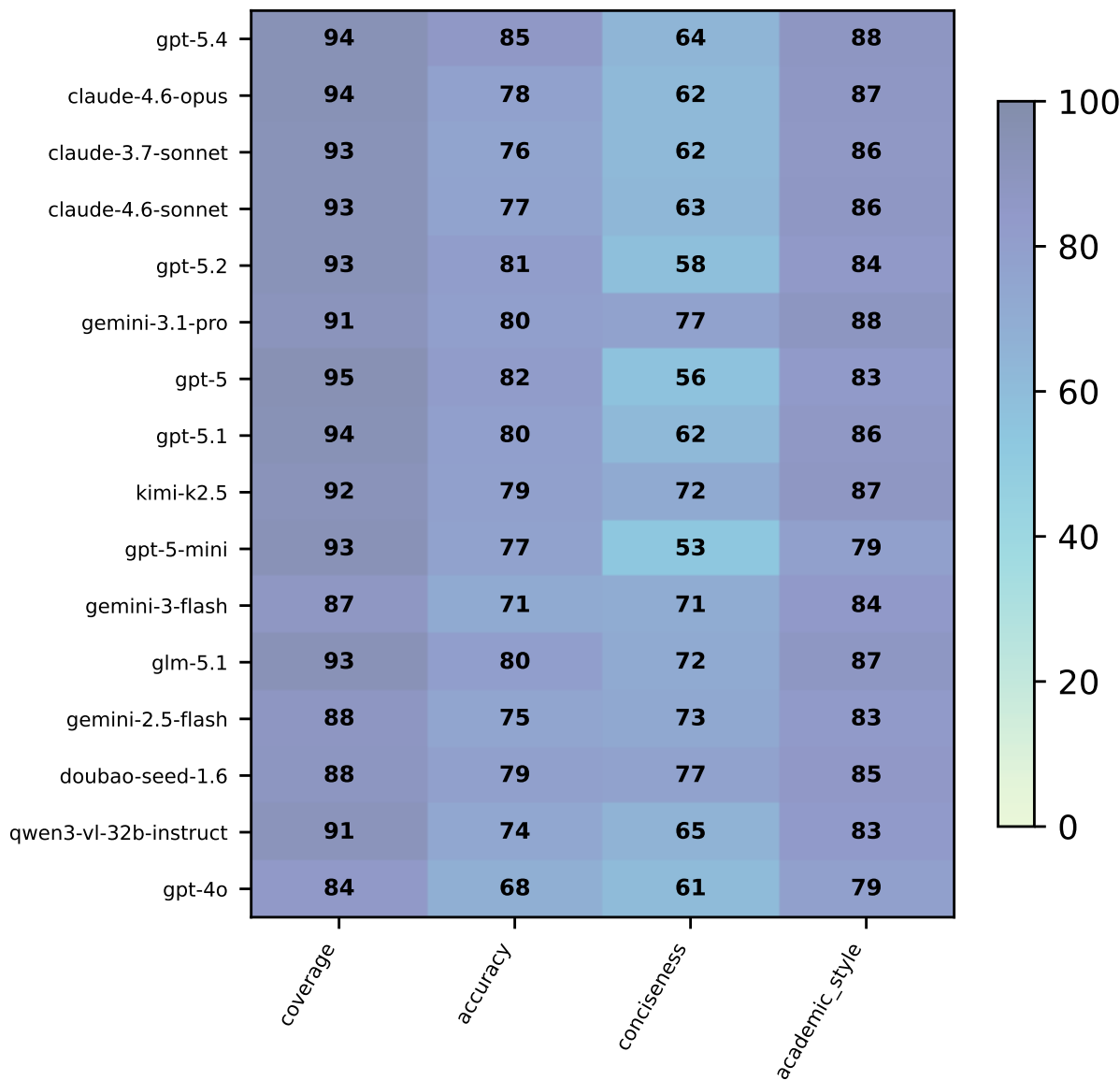


Figure 10: Performance breakdown across Paper Abstraction (PA) dimensions: Coverage (Cov), Accuracy (Acc), Conciseness (Conc), and Academic Style (Style). Values represent percentage scores (0–100).

Paper Abstraction tasks require models to summarize and analyze economic research papers across four dimensions: coverage, accuracy, conciseness, and academic style. Figure 10 shows that gemini-3-flash leads this task (77.0%), with particularly strong performance in coverage and accuracy dimensions.

Coverage scores, measuring whether summaries capture all key contributions, show the widest performance spread among dimensions. Top models (gpt-5, gpt-5.4, claude-4.6-opus, gpt-5.1, glm-5.1) achieve 91–95% coverage, while mid-tier models range from 86–92%. This suggests that identifying salient information from lengthy academic papers requires sophisticated comprehension capabilities.

Accuracy scores are generally high across all models, indicating that when models do extract information, they rarely introduce factual errors. However, conciseness proves challenging—many models produce verbose summaries that include excessive detail. The heatmap

reveals that Claude models (claude-4.6-opus, claude-3.7-sonnet, claude-4.6-sonnet) show particularly balanced performance across all four dimensions, suggesting well-calibrated summarization capabilities.

Academic style scores, evaluating whether summaries use appropriate scholarly language and structure, correlate strongly with overall model capability. Frontier models consistently score above 80%, while the oldest model in the cohort (gpt-4o) trails at 78.6%, indicating that academic writing conventions require substantial model capacity to master.

G.4 Literature Review (LR)

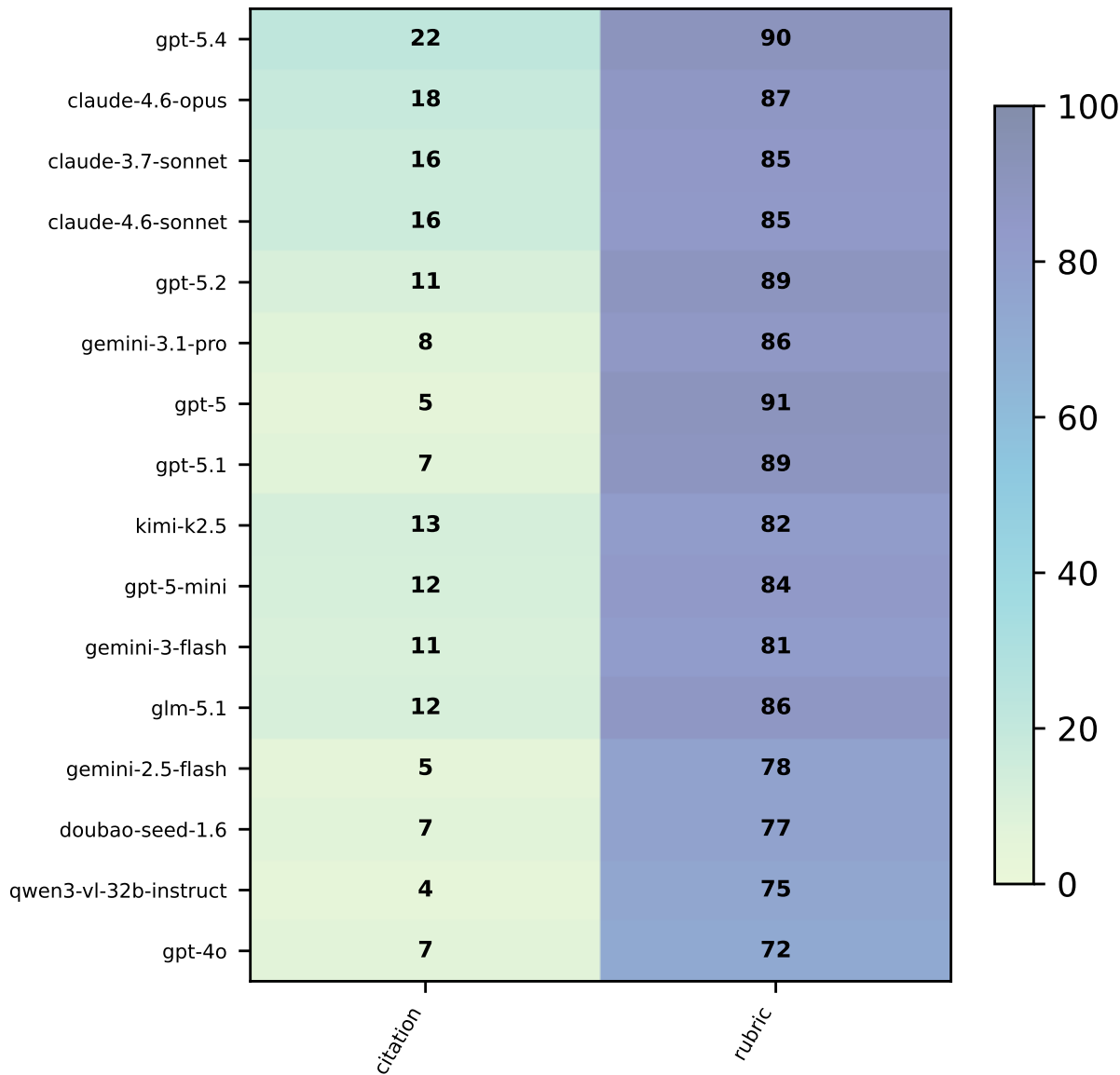


Figure 11: Performance breakdown across Literature Review (LR) dimensions: Citation Score (Cite) and Rubric Score (Rubric). Values represent percentage scores (0–100).

Literature Review is the most challenging task in AER-bench, with the highest-performing model (gpt-5.4) achieving only 56.0%. Figure 11 decomposes performance into citation score (accuracy of cited papers) and rubric score (quality of synthesis and analysis). The heatmap reveals a stark contrast: citation scores are extremely low across models, while rubric scores are substantially higher.

Citation scores measure whether models correctly identify and cite relevant papers from a provided reference list. The extremely low scores indicate that models struggle with precise bibliographic matching and relevance assessment. Even the best-performing model, gpt-5.4, achieves only 21.6% citation accuracy, followed by claude-4.6-opus at 18.3%. Most models score below 15%, suggesting that fine-grained citation retrieval and bibliographic matching remain a fundamental weakness for current LLMs.

Rubric scores, evaluating the quality of literature synthesis, thematic organization, and critical analysis, are substantially higher. Top models achieve 85–90% rubric scores (e.g., gpt-5 at 90.5%, gpt-5.4 at 90.4%), indicating that models can produce well-structured literature review prose when the task does not require precise citation matching. The stark contrast between citation and rubric scores reveals that the primary bottleneck in literature review is not analytical writing quality but rather the ability to accurately identify and attribute relevant sources.

Interestingly, the performance gap between frontier and mid-tier models is smaller in LR than in other tasks. This suggests that literature review capabilities scale less predictably with model size and training, possibly because the task requires specialized academic reasoning patterns underrepresented in general training data.

G.5 Model Generation (MG)

Model Generation tasks evaluate models' ability to construct formal economic models from natural language scenarios, specifying economic agents, optimization problems, constraints, and equilibrium or game mechanisms. Figure 12 shows performance across element-quality categories: the share of rubric elements judged full, substantial, or partial.

gpt-5.4 leads this task (85.8%), followed closely by gpt-5 (84.2%) and gpt-5.1 (82.8%). The heatmap reveals that top models produce predominantly full-quality elements (60–80% shares), indicating strong model-construction reasoning. However, certain items—particularly those requiring creative problem-solving or detailed economic modeling—show greater performance variation.

Claude models demonstrate balanced performance across rubric elements, with claude-3.7-sonnet achieving 81.7%. Gemini models show more uneven performance, with gemini-3.1-pro scoring only 70.0% overall despite strong performance in other tasks. This suggests that model generation requires different capabilities than comprehension or analysis tasks.

Lower-tier models (doubao-seed-1.6, qwen3-v1-32b-instruct) score below 65%, indicating that constructing complete economic models requires substantial reasoning capacity. The task demands understanding of economic agents, optimization problems, constraints, and equilibrium concepts—knowledge that appears concentrated in frontier models.

G.6 Economic Analysis (EA)

Economic Analysis tasks require models to interpret regression results, explain economic mechanisms, connect findings to theory, and discuss policy implications. Figure 13 shows performance across six dimensions, revealing substantial variation in model capabilities.

gpt-5.4 achieves the highest overall EA score (81.6%), with particularly strong performance in accuracy and writing quality dimensions. The heatmap shows that accuracy scores—measuring correct interpretation of statistical results—range from 65–85% across models, with frontier models (gpt-5.4 at 84.1%, gpt-5.2 at 82.3%, gpt-5-mini at 80.5%) scoring above 80%, indicating solid quantitative reasoning. However, deeper analytical dimensions show more variation.

Economic interpretation and mechanism explanation prove moderately challenging, with most models scoring 70–85%. These dimensions require understanding causal pathways and economic logic beyond statistical patterns. Theory connection is the most challenging EA dimension, with scores ranging from 56–73%. Even top models like gpt-5.4 achieve only 73.3% on theory connection, suggesting that linking empirical findings to established theoretical frameworks requires sophisticated domain knowledge.

Policy implication scores are slightly higher than theory connection (ranging 63–76%), but still represent a significant challenge. This dimension requires extrapolating from research findings to real-world applications, considering practical constraints and normative judgments.

Writing quality scores are uniformly high (>75%) across frontier models, indicating that models can produce well-structured, coherent analytical text. The strong correlation between writing quality and overall EA performance suggests that clear communication and analytical depth are closely linked capabilities.

G.7 Concept Identification (CI)

Concept Identification tasks evaluate models' ability to identify and classify economic concepts from historical AER papers, assessing understanding across eight dimensions. Figure 14 reveals that this task shows the highest performance variance across models, with claude-3.7-sonnet leading (73.0%) while gpt-5 scores only 46.9%.

The heatmap shows that Claude models (claude-4.6-opus, claude-3.7-sonnet, claude-4.6-sonnet) consistently outperform other providers on CI tasks, achieving 72.7–73.0% overall scores. However, their performance varies dramatically across individual dimensions, ranging from as low as 10% (time horizon for claude-4.6-sonnet) to as high as 86.2% (uniqueness for claude-3.7-sonnet). This suggests that Anthropic's training or architecture may be particularly well-suited for certain aspects of conceptual classification and historical document understanding, while struggling with temporal reasoning dimensions.

Relevance scores are generally the highest across models (ranging 58–87%), indicating that identifying whether a concept is economically relevant is relatively straightforward. However, other dimensions show much greater variance and difficulty. Math formalization scores range

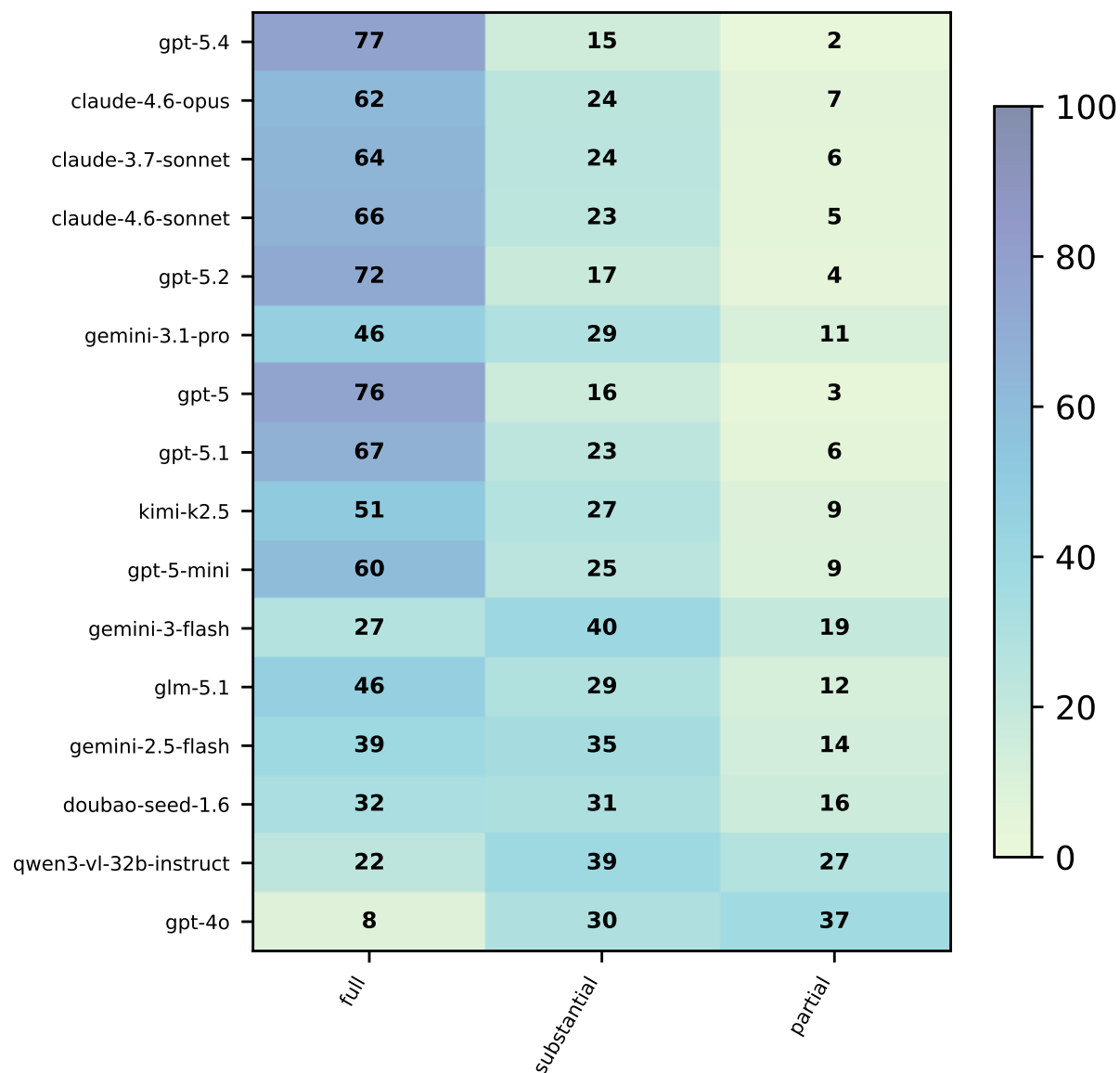


Figure 12: Performance breakdown across Model Generation (MG) element-quality categories: the share of rubric elements attaining each quality level (full, substantial, partial). Values represent percentage scores (0–100).

from 18.8% to 85.0%, with some models excelling (doubao-seed-1.6: 85.0%, gpt-5: 81.2%) while others struggle (qwen3-vl-32b-instruct: 18.8%). More nuanced dimensions—uniqueness, stability, existence—also prove challenging, with uniqueness showing extreme variance (1.2–95.0%) across models.

The equilibrium scene, time characteristic, and time horizon dimensions show the greatest performance spread. These require understanding subtle distinctions in how economic concepts are defined and applied across different historical contexts. The low scores suggest that temporal reasoning and historical concept evolution are particularly difficult for current foundation models.

Interestingly, GPT-5 series models show unusually weak CI performance relative to their strong performance on other tasks. This may indicate that the training data or optimization objectives for these models prioritize contemporary language patterns over historical economic terminology, creating a systematic weakness in this dimension.

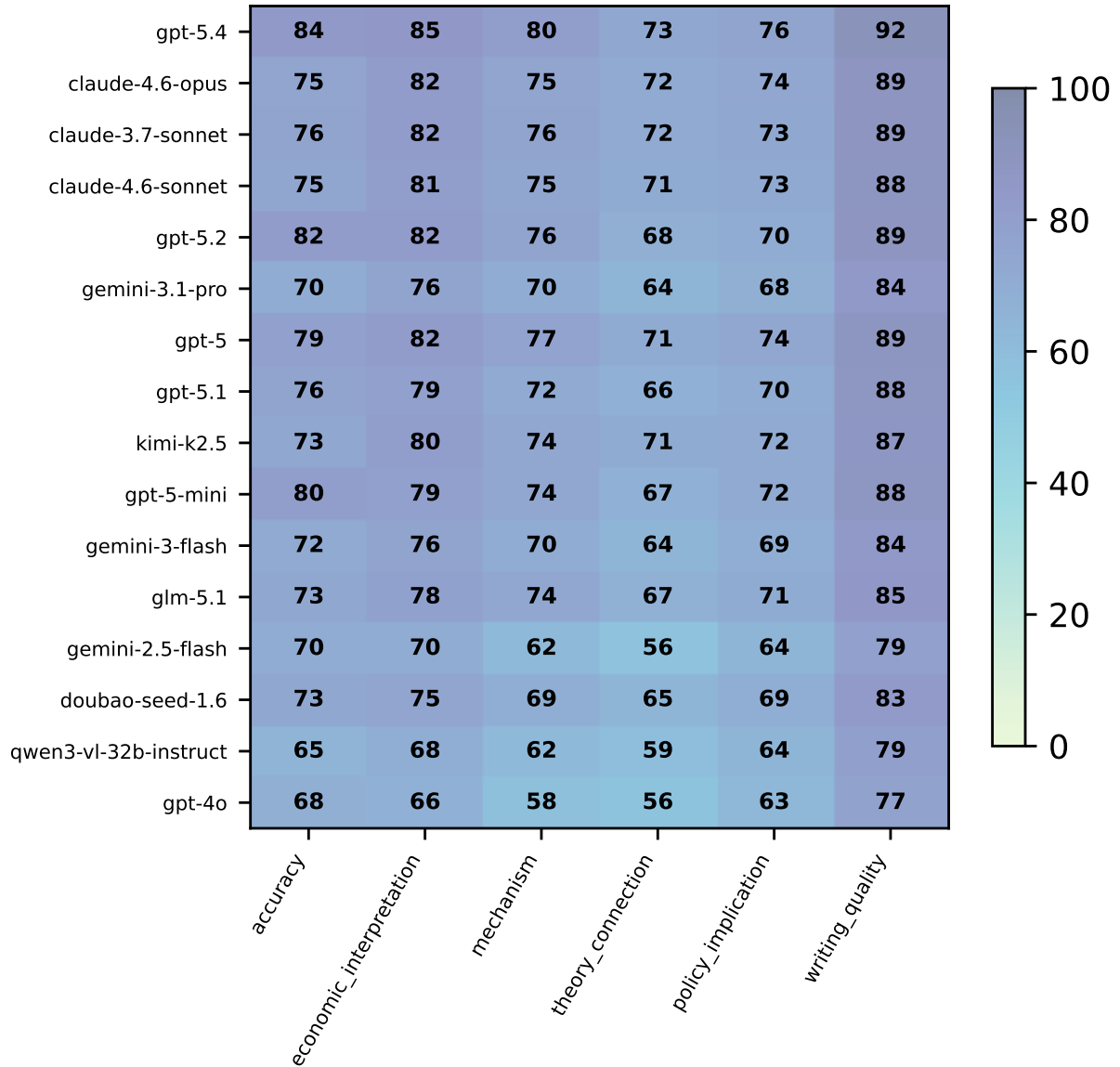


Figure 13: Performance breakdown across Economic Analysis (EA) dimensions: Accuracy (Acc), Economic Interpretation (Interp), Mechanism (Mech), Theory Connection (Theory), Policy Implication (Policy), and Writing Quality (Write). Values represent percentage scores (0–100).

H Task Generation Methodology

This appendix provides comprehensive documentation of the multi-round generation pipeline used to construct evaluation items for each task dimension in AER-bench. All tasks except Concept Identification (CI) employ LLM-based generation with carefully designed prompts and validation steps. CI items are manually annotated from historical AER papers and are therefore not covered in this appendix.

For each task, we describe: (1) the complete multi-round generation workflow, (2) key design rationale, (3) full prompt templates with all critical instructions, (4) complete output format specifications, and (5) scoring rules and evaluation criteria. All generation uses gpt-5.4 as the extraction model.

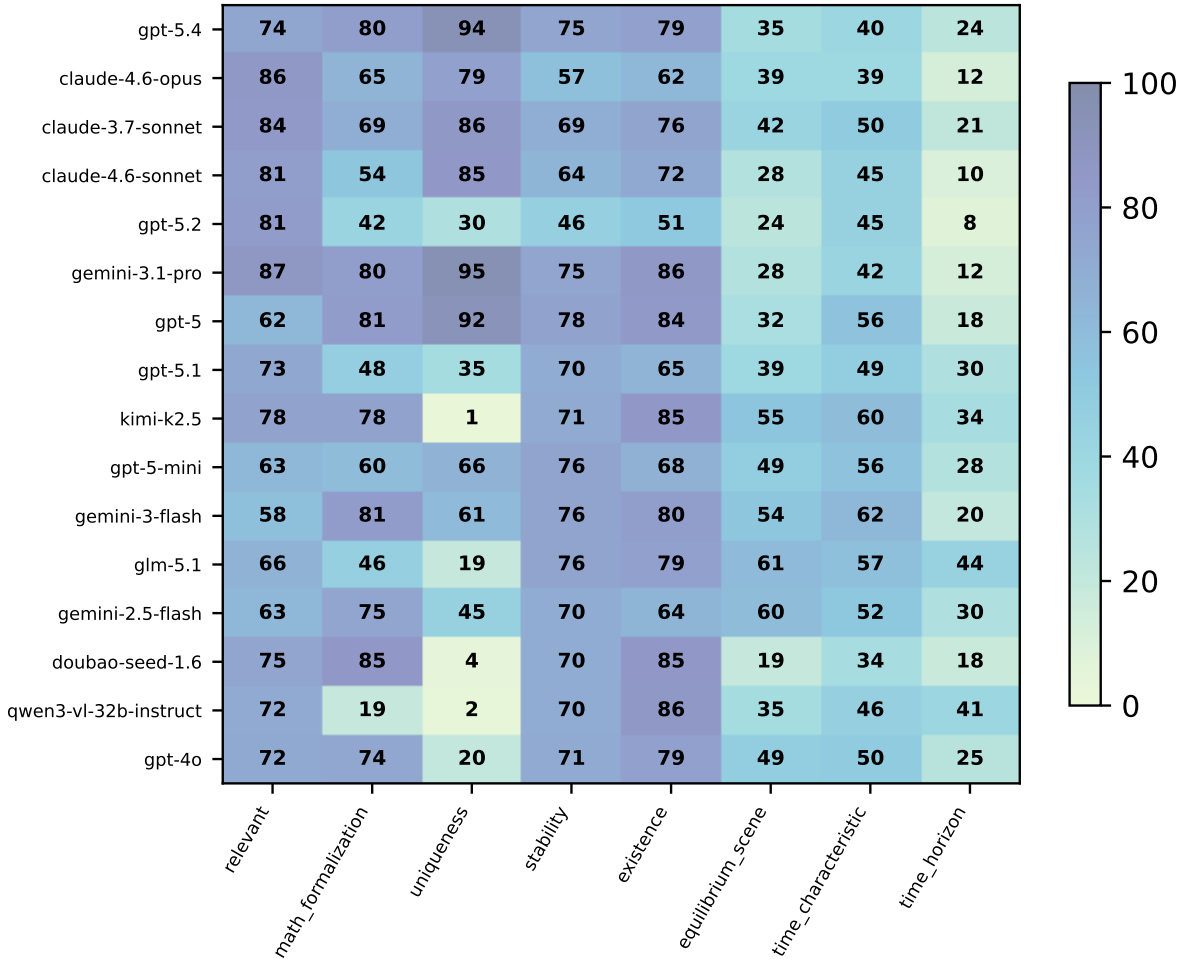


Figure 14: Performance breakdown across Concept Identification (CI) dimensions: Relevant, Math Formalization, Uniqueness, Stability, Existence, Equilibrium Scene, Time Characteristic, and Time Horizon. Values represent percentage scores (0–100).

H.1 Economic Knowledge (EK)

H.1.1 Task Overview. Economic Knowledge (EK) is a Level 1 (foundational capability) task that evaluates models’ depth of understanding of economic literature. The core evaluation goal is to test whether a model can answer questions about specific papers’ model setups, core findings, and methodological choices. Each paper produces 20 objective questions: 5 single-choice, 5 true/false, 5 fill-in-blank, and 5 ordering questions. The task depends only on PDF input (no supplementary materials required) and uses a 2-round generation pipeline.

H.1.2 Generation Workflow. EK employs a two-round pipeline to ensure balanced knowledge coverage and question diversity:

- (1) **Round 1 (PDF input):** Extract 12–15 paper-specific knowledge points spanning six categories: model setup, findings, methods, mechanisms, contributions, and institutional context. Each knowledge point is tagged as *quantitative* (involving specific numbers) or *conceptual* (involving qualitative understanding), with a target of at least 50% conceptual points to avoid over-extraction of numerical facts.
- (2) **Round 2 (text input):** Based on the structured knowledge point list from Round 1, generate 20 objective questions: 5 single-choice, 5 true/false, 5 fill-in-blank, and 5 ordering questions. Each question references the source knowledge point and includes difficulty tags.

H.1.3 Design Rationale. Separating knowledge extraction from question generation prevents two common failure modes: (1) knowledge point omission when simultaneously reading the paper and generating 20 questions, and (2) uneven distribution across question types or content categories. The structured knowledge point list serves as a checklist to ensure comprehensive coverage.

Critical Design Constraint. All questions are designed to test whether an AI system or economics researcher **knows this specific paper**. Questions provide ONLY the first author’s surname + year + a short topic phrase (e.g., “Autor et al. (2013), a paper on Chinese import competition and U.S. labor markets”). The full paper title, journal name, and co-author details do NOT appear in the stem. The answer must require knowledge of the paper’s specific details that cannot be guessed from the topic phrase alone.

H.1.4 Round 1 Prompt Template.

You are a professor of economics. Please read the attached PDF paper in full and extract ****paper-specific knowledge points**** that can be used for exam questions.

Critical design constraint

The exam questions you will later create are designed to test whether an AI system or an economics researcher ****knows this specific paper****. Therefore:

- ****ALL knowledge points must be about this paper's specific content****: its model setup, assumptions, variables, findings, methodology, mechanisms, contributions, or institutional context.
- The questions will provide ONLY the ****first author's surname + year + a short topic phrase**** (e.g., "Autor et al. (2013), a paper on Chinese import competition and U.S. labor markets"). The full paper title, journal name, and co-author details will NOT appear in the stem. The ****answer must require knowledge of the paper's specific details**** that cannot be guessed from the topic phrase alone.
- A well-read economist who has studied this paper (or can search for it) should be able to answer; one who has not should find it difficult.

Knowledge point categories

Extract knowledge points spanning as many of the following categories as the paper supports:

- **`model\_setup`**: Specific assumptions, agent definitions, variable choices, functional forms, or information structures in the paper's model
- **`finding`**: Core empirical or theoretical results -- coefficient directions, magnitudes, significance levels, comparative statics, or proposition statements
- **`method`**: Identification strategy, estimation approach, instrument choice, data sources, or sample construction decisions and their rationale
- **`mechanism`**: Economic mechanisms or causal channels that the paper identifies, tests, or discusses
- **`contribution`**: The paper's stated contributions, how it differs from or extends prior work
- **`institution`**: Institutional background, policy details, or market features specific to the paper's setting

CRITICAL: Content type balance

You MUST extract a balanced mix of ****quantitative**** and ****conceptual**** knowledge points:

- ****Quantitative**** knowledge points: involve specific numbers, coefficients, percentages, counts, or magnitudes from the paper (e.g., "The estimated effect is -0.37", "The sample includes 126 DMAs")
- ****Conceptual**** knowledge points: involve model assumptions, mechanism descriptions, methodology choices, variable definitions, institutional features, or qualitative findings (e.g., "The paper uses a shift-share instrument", "The model assumes risk-neutral agents")

****Target**: at least 50% of knowledge points must be conceptual. Do not over-extract numerical facts at the expense of conceptual understanding.

H.1.5 Round 1 Output Format.

```
{ "paper_id": "<paper ID>", "paper_title": "<paper title>", "knowledge_points": [ { "kp_id": "KP01",
"category": "model_setup | finding | method | mechanism | contribution | institution",
"content_type": "quantitative | conceptual", "statement": "<specific knowledge point>",
"paper_context": "<2-3 sentence description stating what aspect of the paper this addresses, the
specific detail tested, and where it appears>" } ] }
```

H.1.6 Round 2 Prompt Template.

You are a professor of economics. Below is a list of paper-specific knowledge points extracted from an economics paper. Based on these knowledge points, generate ****20 objective questions**** to test whether an examinee knows this specific paper.

Question type requirements

Generate exactly:

- 5 single-choice questions (4 options each, labeled A/B/C/D)
- 5 true/false questions
- 5 fill-in-the-blank questions
- 5 ordering questions (4 items to order)

CRITICAL: Content type balance

Ensure that across all 20 questions, at least 50% test **conceptual** knowledge (not just numerical recall). Use the `content_type` tags from the knowledge points to guide this balance.

Mandatory stem format

Every question stem **MUST** begin with the paper reference in this exact format:

"In [FirstAuthor] et al. ([Year]), a paper on [brief topic phrase], ..."

Example: "In Autor et al. (2013), a paper on Chinese import competition and U.S. labor markets, what identification strategy is used?"

Do NOT include:

- Full paper title
- Journal name
- Full author list beyond first author

Question design principles

1. **Paper-specific**: Every question must require knowledge of this specific paper's content. Avoid generic economics questions.
2. **Unambiguous**: Each question must have a single, clearly correct answer based on the paper.
3. **Difficulty distribution**: Aim for a mix of medium (60%) and hard (40%) questions. Easy questions that can be guessed should be avoided.
4. **Source tracking**: Each question must reference its source knowledge point via `source_kp` field.

Single-choice question requirements

- Provide exactly 4 options (A, B, C, D)
- All distractors must be plausible but clearly incorrect based on the paper
- Avoid "all of the above" or "none of the above" options
- Distractors should not be trivially eliminable

True/false question requirements

- Statement must be specific enough that truth value is unambiguous
- Avoid statements that are "technically true but misleading" or vice versa
- Explanation must cite specific paper content

Fill-in-the-blank question requirements

- Blank should test a specific term, number, or concept
- Use "___" to mark the blank position
- Provide canonical answer and list of acceptable synonyms/variants
- Acceptable answers should account for: capitalization differences, singular/plural forms, common abbreviations

Ordering question requirements

- Provide exactly 4 items to be ordered
- Specify the ordering criterion clearly in the stem (e.g., "from smallest to largest effect size",

"in chronological order of implementation")
 - Items must have an unambiguous correct ordering based on the paper
 - Answer field contains the items in correct order

H.1.7 Round 2 Output Format.

```
{ "paper_id": "<paper ID>", "questions": [ { "id": "EK_<paper_id>_SC01", "type": "single_choice",
"content_type": "conceptual | quantitative", "source_kp": "KP03", "difficulty": "medium | hard",
"knowledge_tags": ["<knowledge category>"], "stem": "<question stem starting with paper reference>",
"options": { "A": "<option A>", "B": "<option B>", "C": "<option C>", "D": "<option D>" }, "answer":
"B", "explanation": "<explanation referencing specific paper content>" }, { "id":
"EK_<paper_id>_TF01", "type": "true_false", "content_type": "conceptual | quantitative",
"source_kp": "KP07", "difficulty": "medium | hard", "knowledge_tags": ["<knowledge category>"],
"stem": "<statement about the paper to judge true/false>", "answer": true, "explanation":
"<explanation referencing specific paper content>" }, { "id": "EK_<paper_id>_FB01", "type":
"fill_in_blank", "content_type": "conceptual | quantitative", "source_kp": "KP05", "difficulty":
"medium | hard", "knowledge_tags": ["<knowledge category>"], "stem": "<fill-in stem with ___
referencing paper content>", "answer": "<canonical answer>", "acceptable_answers": ["<synonym 1>",
"<synonym 2>"], "explanation": "<explanation referencing specific paper content>" }, { "id":
"EK_<paper_id>_OD01", "type": "ordering", "content_type": "conceptual | quantitative", "source_kp":
"KP08", "difficulty": "medium | hard", "knowledge_tags": ["<knowledge category>"], "stem":
"<ordering stem stating the criterion, referencing paper>", "items": ["<item A>", "<item B>", "<item
C>", "<item D>"], "answer": ["<item B>", "<item A>", "<item D>", "<item C>"], "explanation":
"<explanation referencing specific paper content>" } ] }
```

Question Type	Fully Correct	Partially Correct	Incorrect
Single Choice (5 questions)	1 point	–	0 points
True/False (5 questions)	1 point	–	0 points
Fill-in-Blank (5 questions)	1 point (case-insensitive, or in acceptable_answers)	–	0 points
Ordering (5 questions)	1 point (exact match)	0.5 points (Kendall $\tau \geq 0.5$)	0 points

H.1.8 *Scoring Rules.* Total score = sum of individual question scores / number of questions.

H.2 Mathematical Modeling (MM)

H.2.1 *Task Overview.* Mathematical Modeling (MM) is a Level 1 (foundational capability) task that evaluates models' ability to write reasonable model equations, constraints, or complete key derivation steps given an economic scenario. The core evaluation goal is to test formula-level mathematical skills. Each paper produces 3 questions spanning three types: equation design, formula derivation, and equilibrium solving. The task depends only on PDF input and uses a 2-round generation pipeline.

H.2.2 *Generation Workflow.* MM uses a two-round pipeline to systematically extract and select mathematical content:

- (1) **Round 1 (PDF input):** Systematically extract all mathematical modeling content from the paper, including model setup (agents, variables, parameters), key equations (objective functions, constraints, FOCs), propositions and proofs, and derivation chains. For empirical papers, extract regression equations and identification assumptions. Output includes a `suitable_for_mm` flag.
- (2) **Round 2 (text input):** Based on Round 1's structured mathematical content, design 3 self-contained mathematical modeling questions spanning three types: equation design, formula derivation, and equilibrium solving. Each question includes scenario description, notation conventions, known conditions, and standard answer in LaTeX.

H.2.3 *Design Rationale.* Economic papers scatter mathematical content across main text, appendices, and footnotes. Round 1 performs exhaustive extraction and builds derivation chains, allowing Round 2 to select the most suitable elements for exam questions without missing key formulas or choosing overly trivial derivations.

LaTeX Escaping in JSON. Since output is JSON containing LaTeX strings, all backslashes must be escaped: $\frac{a}{b}$ becomes `\frac{a}{b}` in JSON. Every backslash `\` in LaTeX must be written as `\\` in the JSON string. Double backslashes in LaTeX (e.g., `\\` for line breaks) must be `\\\\` in JSON.

H.2.4 Round 1 Prompt Template.

You are an expert in mathematical economics. Read the full attached PDF paper and systematically extract the paper's mathematical modeling content.

Extraction requirements

Identify and record the following:

1. ****Model setup****: Definitions of all economic agents, variables, and parameters
2. ****Key equations****: Objective functions, constraints, equilibrium conditions, first-order conditions, etc.
3. ****Propositions and proofs****: Propositions, Theorems, and Lemmas in the paper and their core derivation steps
4. ****Derivation chains****: Logical relations between formulas (which formula follows from which prior conditions)

Handling non-theoretical papers

If the paper is empirical:

- Extract regression / estimation equations (e.g. DID, IV, structural models, etc.)
- Extract mathematical formulations of identification assumptions
- Extract statistical or asymptotic properties of the model

If the paper contains almost no mathematical derivations suitable for exam items (e.g. only reports regression results, lacking reconstructible equations and derivation chains):

- Set ``suitable_for_mm = false``
- Provide ``reason_not_suitable``
- Do not invent mathematical models or derivation chains

LaTeX in JSON -- escaping rules

Since the output is JSON containing LaTeX strings, you MUST follow these escaping rules:

- Every backslash ``\`` in LaTeX must be written as ``\\`` in the JSON string
- For example: ``\frac{a}{b}`` must be written as ``\\frac{a}{b}``
- Newlines in LaTeX should be ``\\n`` in the JSON string
- Double backslashes in LaTeX (e.g., ``\\`` for line breaks) must be ``\\\\`` in JSON
- Test: the JSON must be parseable by a standard JSON parser

H.2.5 Round 1 Output Format.

```
{ "paper_id": "<paper ID>", "paper_title": "<paper title>", "suitable_for_mm": true,
  "reason_not_suitable": "", "paper_type": "theoretical | empirical | mixed", "model_elements": {
    "agents": [ { "name": "<agent>", "variables": [ "<variable (LaTeX)>" ], "parameters": [ "<parameter>" ] },
    "key_equations": [ { "eq_id": "EQ01", "label": "<equation label, e.g. objective function, budget constraint, FOC>", "latex": "<LaTeX formula>", "context": "<where the equation appears and its meaning (1 sentence)>", "depends_on": [ "EQ00" ] },
    "propositions": [ { "prop_id": "P01", "statement": "<verbal statement of the proposition>", "key_result_latex": "<LaTeX expression of the core result>", "proof_sketch": "<key proof steps (2-3 steps)>", "depends_on": [ "EQ01", "EQ03" ] },
    "candidate_questions": [ { "suggested_type": "equation_design | derivation | equilibrium_solving", "target_element": "EQ03 | P01", "why_suitable": "<why this element is suitable as an exam item (1 sentence)>", "estimated_difficulty": "medium | hard" } ] } }
```

H.2.6 Round 2 Prompt Template.

You are an expert in writing mathematical economics exam questions. Below is extracted mathematical modeling content from an economics paper. Based on this content, design ****3 mathematical modeling questions****.

Question type requirements

Design exactly 3 questions, one from each type:

1. ****Equation design****: Given a scenario, write the appropriate objective function, constraint, or equilibrium condition
2. ****Formula derivation****: Derive a key result from given premises
3. ****Equilibrium solving****: Characterize equilibrium or solve for optimal choices

Self-containment principle

Each question must be **completely self-contained**:

- Provide all necessary background and notation in the question stem
- Define all variables and parameters used
- State all assumptions and known conditions
- The examinee should NOT need to refer to the original paper

Question structure

Each question should include:

1. **Scenario description**: Economic context and setup (2–3 sentences)
2. **Notation conventions**: List of all symbols with their meanings
3. **Known conditions**: What is given or assumed
4. **Task**: What the examinee needs to derive or solve
5. **Standard answer**: Complete solution in LaTeX

LaTeX escaping reminder

Remember: every ``\`` in LaTeX must be ```\`` in JSON output.

Difficulty calibration

- **Medium**: Requires applying standard techniques (FOC, envelope theorem, budget constraint substitution)
- **Hard**: Requires multi-step reasoning, non-obvious substitutions, or careful handling of corner cases

Target: 1–2 medium questions, 1–2 hard questions.

H.2.7 Round 2 Output Format.

```
{ "paper_id": "<paper ID>", "questions": [ { "id": "MM_<paper_id>_Q01", "type": "equation_design |
derivation | equilibrium_solving", "difficulty": "medium | hard", "scenario": "<economic scenario
description (2-3 sentences)>", "notation": { "<symbol>": "<meaning>", "x": "consumption quantity",
"p": "price" }, "known_conditions": [ "<condition 1>", "<condition 2>" ], "task": "<what the
examinee needs to do>", "standard_answer": { "latex": "<complete LaTeX solution>", "explanation":
"<step-by-step explanation>" }, "rubric": { "correctness": "<what constitutes a correct answer>",
"key_elements": [ "<element 1>", "<element 2>" ] } ] }
```

H.2.8 Scoring Method. MM uses a hybrid scoring approach combining structured comparison and LLM-as-Judge:

Scoring Dimensions (0–10 points each, total 40).

- (1) **Formula correctness** (0–10): All formulas are mathematically equivalent to the reference; derivation steps are valid. 9–10: fully correct; 6–8: minor errors but core structure correct; 3–5: significant errors but partial understanding evident; 0–2: fundamentally incorrect.
- (2) **Notation consistency** (0–10): All symbols match the problem setup and are used consistently. 9–10: perfect consistency; 6–8: minor inconsistencies; 3–5: some symbols undefined or inconsistent; 0–2: major notation confusion.
- (3) **Derivation completeness** (0–10): All intermediate steps shown; a reader can verify each step. 9–10: all steps present; 6–8: key steps present but some algebra omitted; 3–5: major steps missing; 0–2: only final answer or fragmentary.
- (4) **Economic intuition** (0–10): Clear interpretation of results in economic terms; explains what the math means. 9–10: excellent economic interpretation; 6–8: some interpretation provided; 3–5: minimal economic context; 0–2: no economic interpretation.

Normalized score = total / 40.

H.3 Paper Abstraction (PA)

H.3.1 Task Overview. Paper Abstraction (PA) is a Level 2 (comprehensive capability) task that evaluates models' ability to write abstracts after reading condensed body text. The core evaluation goal is to test whether generated abstracts cover key points from the original abstract. Each paper produces 1 evaluation item containing condensed body text (approximately 30% of original length, excluding abstract) and 5–10 atomic semantic propositions as evaluation criteria. The task depends only on PDF input and uses a 3-round generation pipeline.

H.3.2 Generation Workflow. PA employs a three-round pipeline to produce condensed text and atomic evaluation criteria:

- (1) **Round 1 (PDF input)**: Extract the original abstract verbatim, identify section structure, and summarize core arguments for each major section (1–2 sentences per section). Also extract 2–4 claimed contributions.

- (2) **Round 2 (PDF + Round 1)**: Generate condensed body text at approximately 30% of the original length, excluding the abstract. Use Round 1’s section structure to guide selective retention of core arguments, key findings, and methodological details while removing redundant exposition.
- (3) **Round 3 (text input)**: Decompose the original abstract into 5–10 atomic semantic propositions (key points) that serve as evaluation criteria. Cross-validate these propositions against Round 1’s section arguments to ensure completeness.

H.3.3 Design Rationale. The three-round separation ensures: (1) Round 1 provides structural guidance for Round 2’s condensation, (2) Round 2 can refer back to the PDF to avoid incorrect deletions of critical arguments, and (3) Round 3 focuses purely on atomic proposition decomposition without needing PDF access.

Examinee Input. Examinees receive only the condensed body text (approximately 30% of original length, excluding abstract) in plain text format. This design reduces token cost, supports models without PDF input, and focuses evaluation on “distilling abstracts from core information” rather than “processing long documents.”

H.3.4 Round 1 Prompt Template.

You are an expert in economics research methods. Please read the attached PDF paper and complete the following information extraction.

Task 1: Extract the original abstract

Locate and extract the paper's abstract text in full (verbatim copy; do not paraphrase).

Task 2: Paper structure and core arguments by section

Identify the paper's section structure and summarize the core argument in 1-2 sentences for each major section.

Task 3: Identify the paper's core contributions

List 2-4 main contributions the paper explicitly claims (typically at the end of the introduction or in the conclusion).

H.3.5 Round 1 Output Format.

```
{ "paper_id": "<paper ID>", "paper_title": "<paper title>", "original_abstract": "<full original abstract, verbatim>", "sections": [ { "section_number": "1", "section_title": "<section title>", "core_argument": "<core argument (1-2 sentences)>" } ], "claimed_contributions": [ "<contribution 1>", "<contribution 2>" ], "methodology_summary": "<one-sentence overview of methods>", "key_findings_summary": "<one-sentence overview of findings>" }
```

H.3.6 Round 2 Prompt Template.

You are an academic editor. Below is the section structure extracted from an economics paper. Please re-read the attached PDF source and compress the body of the paper into a **condensed version** of approximately 30% of the original length.

Compression goal

Produce a **plain-text condensed body** whose length is approximately 30% of the original paper body (excluding references and appendices). This text will be the sole input for examinees, who will write an abstract based on it.

Compression principles

1. **Must not include the original abstract** -- that is what examinees need to produce
2. **Preserve the paper's core argumentative chain**: research question -> methods -> key findings -> conclusions
3. **Organize in section order**, retaining the 2-4 most critical sentences of core argument per section
4. **Retain key data and quantitative results** (e.g., core regression coefficients, effect sizes, statistical significance)
5. **Retain core methodological description** (e.g., identification strategy, model specification, data sources)
6. **Omit**: detailed literature review, secondary variants of robustness checks, appendix material, redundant exposition

7. **Do not paraphrase** -- prefer direct quotations or minimally shortened phrases from the original
8. Use **English** and maintain an academic register
9. Prefix each section paragraph with `[Section N: Title]` for clarity

H.3.7 Round 3 Prompt Template.

You are an expert in academic writing. Below is the original abstract of an economics paper. Decompose it into **5-10 atomic semantic propositions** (key points) that will serve as evaluation criteria for examinee-generated abstracts.

Decomposition principles

1. **Atomic**: Each key point should express ONE distinct claim or fact
2. **Verifiable**: Each point should be checkable against candidate text
3. **Complete coverage**: Together, the points should cover all major elements of the original abstract
4. **Category balance**: Include points from multiple categories: research question, method, finding, contribution

Key point categories

- ``research_question``: What economic question the paper addresses
- ``method``: Identification strategy, data, or modeling approach
- ``finding``: Core empirical or theoretical results
- ``contribution``: How the paper advances the literature

must_appear flag

Mark a key point as ``must_appear: true`` if it represents a core contribution or finding that **MUST** be mentioned in any reasonable abstract of this paper. Mark as ``false`` if it is a supporting detail that could reasonably be omitted in a concise abstract.

Target: 60-70% of key points should be `must_appear: true``.

H.3.8 Round 3 Output Format.

```
{ "paper_id": "<paper ID>", "key_points": [ { "kp_id": "KP01", "statement": "<atomic proposition>",
"category": "research_question | method | finding | contribution", "must_appear": true } ] }
```

H.3.9 Scoring Method. PA uses LLM-as-Judge evaluation across four dimensions (0-10 points each):

- (1) **Coverage** (0-10): Are all or nearly all reference key points mentioned or clearly implied? 9-10: all key points covered; 6-8: most covered with 1-2 minor omissions; 3-5: several important points missing; 0-2: only 1-2 points covered.
- (2) **Accuracy** (0-10): Are all stated claims factually correct per the paper content? 9-10: all correct; 6-8: minor inaccuracies; 3-5: some factual errors; 0-2: major errors.
- (3) **Conciseness** (0-10): Is the abstract within word limit with no redundancy? 9-10: within limit, every sentence adds value; 6-8: mostly concise with minor redundancy; 3-5: noticeably wordy or exceeds limit; 0-2: excessively long.
- (4) **Academic style** (0-10): Does it read like a published abstract in a top economics journal? 9-10: publication-ready; 6-8: generally academic with some informal phrasing; 3-5: mixed register; 0-2: not academic.

H.4 Literature Review (LR)

H.4.1 Task Overview. Literature Review (LR) is a Level 2 (comprehensive capability) task that evaluates models' ability to write introductions with citation networks consistent with the original paper. The core evaluation goal is to test whether models can produce literature reviews that align with the paper's actual citation structure. Each paper produces 1 evaluation item containing condensed body text (approximately 50% of original length, excluding introduction and references), three-tier citation classification, citation role annotations, and a paper-specific writing rubric. The task depends only on PDF input and uses a 4-round generation pipeline.

H.4.2 Generation Workflow. LR uses a four-round pipeline to construct citation-aware evaluation items:

- (1) **Round 1 (PDF input)**: Extract the complete reference list from the paper, including author names, titles, publication venues, and years. Tag each reference by publication type (journal article, book, working paper, etc.).
- (2) **Round 2 (PDF + Round 1)**: Analyze the introduction section: identify introduction boundaries, extract the argumentation structure, and annotate each citation's role in the argument. Classify citations into three tiers based on journal quality and relevance to the paper's core argument.

- (3) **Round 3 (PDF + Round 2)**: Generate condensed body text at approximately 50% of original length, excluding the introduction and references. Use Round 2's introduction boundaries to ensure precise exclusion.
- (4) **Round 4 (text input)**: Construct a paper-specific writing rubric based on the argumentation structure identified in Round 2. The rubric captures the paper's specific rhetorical strategy rather than applying a generic template.

H.4.3 Design Rationale. The four-round separation ensures: (1) clean reference list extraction before citation role annotation, (2) precise introduction exclusion using Round 2 boundaries, and (3) customized rubric construction based on the paper's specific argumentation strategy.

Citation Tier System. Citations in the introduction are classified into three tiers based on journal quality and relevance:

- **Tier 1 (Must-cite)**: Core references indispensable to the argument, typically from top journals (AER, Econometrica, QJE, JPE, REStud) or high-impact sources, but not limited to these—any high-impact work critical to the argument qualifies.
- **Tier 2 (Should-cite)**: Relevant formal academic publications from reputable but non-top journals, academic books, or well-known working papers (NBER, CEPR).
- **Tier 3 (Optional)**: Lower-relevance or less authoritative sources (obscure journals, general preprints, technical reports).

Scoring Formula. Weighted citation recall: $0.8 \times \text{Tier1_recall} + 0.2 \times \text{Tier2_recall}$ (Tier 3 not scored). Final score: 50% citation recall + 50% writing rubric.

H.4.4 Round 1 Prompt Template.

You are an expert in academic reference management. Read the attached PDF paper and extract the ****complete reference list**** from the References / Bibliography section.

Extraction requirements

For each reference, extract:

- Authors (standard format)
- Year
- Title
- Journal / publisher
- Publication type (venue_type)
- A standardized citation_key

venue_type categories

Label each item by its publication source:

- ``top5``: economics Top 5 journals (AER, Econometrica, QJE, JPE, REStud)
- ``top_field``: widely recognized top field journals (e.g., JF, JFE, RFS, JME, JEEA, AEJ series, JHE, JDE, JUE, JPubE, JLE, RAND, EJ, REE, etc.)
- ``good_journal``: well-known but non-top-tier formal journals
- ``book``: scholarly monographs or book chapters
- ``conference``: conference papers
- ``working_paper_top``: working papers from prominent institutions (NBER, CEPR, World Bank, IMF, Fed, etc.)
- ``working_paper_other``: working papers from other institutions
- ``preprint``: preprints (arXiv, SSRN, etc.)
- ``other``: reports, technical documents, news, web pages, etc.

H.4.5 Round 2 Prompt Template.

You are an expert in economics research writing. Re-read the attached PDF paper and analyze the ****introduction section**** in detail.

Task 1: Identify introduction boundaries

Precisely identify where the introduction begins and ends (page numbers and section numbers).

Task 2: Extract argumentation structure

Identify the major argumentative modules in the introduction (e.g., motivation, literature gap, methodology preview, contribution summary). For each module, provide a 2-3 sentence summary.

Task 3: Annotate each citation's role

For every citation that appears in the introduction, record:

- citation_key (matching Round 1 format)
- The argumentative role it plays (e.g., "establishes empirical regularity", "represents prior theoretical approach", "identifies literature gap")
- Tier classification (1/2/3) based on journal quality and relevance

Tier classification criteria

****Tier 1 (Must-cite)**:** Core references indispensable to the argument. Typically from top journals (AER, Econometrica, QJE, JPE, REStud) or high-impact sources, but not limited to these---any high-impact work critical to the argument qualifies.

****Tier 2 (Should-cite)**:** Relevant formal academic publications from reputable but non-top journals, academic books, or well-known working papers (NBER, CEPR).

****Tier 3 (Optional)**:** Lower-relevance or less authoritative sources.

IMPORTANT: Tier 1 and Tier 2 count requirements

After classification, verify:

- Tier 1 should have ≥ 10 citations
- Tier 2 should have ≥ 15 citations

If counts are insufficient, re-examine the introduction and promote relevant citations from lower tiers.

H.4.6 Round 3 Prompt Template.

You are an academic editor. Below is the introduction boundary information extracted from an economics paper. Re-read the attached PDF and compress the body of the paper (EXCLUDING the introduction and references) into approximately 50% of the original length.

Compression goal

Produce plain-text condensed body at ~50% of original length, excluding introduction and references. This will be the examinee's input for writing an introduction.

Compression principles

1. ****Must exclude the entire introduction section**** (use Round 2 boundaries)
2. ****Must exclude the references section****
3. ****Preserve core argumentative chain**:** methods $\rightarrow$ results $\rightarrow$ robustness $\rightarrow$ conclusion
4. ****Retain key quantitative results**** and methodological details
5. ****Omit**:** detailed proofs, secondary robustness checks, appendix material
6. Use section prefixes ``[Section N: Title]`` for clarity

H.4.7 Round 4 Prompt Template.

You are an expert in economics research writing. Based on the introduction analysis from Round 2, construct a ****paper-specific writing rubric**** that captures this paper's actual rhetorical strategy.

Rubric construction principles

1. ****Paper-specific dimensions**:** Identify 4-6 dimensions based on what THIS paper's introduction actually does (not a generic template)
2. ****Concrete evaluation criteria**:** For each dimension, list specific elements to check
3. ****Weight allocation**:** Assign weights summing to 100 based on importance
4. ****Tier-classified citation list**:** Aggregate citation_keys by tier from Round 2

Example dimensions (adapt to paper)

- Motivation and research question clarity
- Literature gap identification
- Methodology preview

- Contribution summary
- Institutional context (if applicable)
- Theoretical framework setup (if applicable)

Output requirements

- writing_rubric: array of dimension objects with name, description, key_elements_to_check, weight
- gold_citations_tiered: object with tier_1, tier_2, tier_3 arrays
- scoring_formula: explicit formula for combining citation and rubric scores

H.4.8 Scoring Method. LR uses dual-track scoring:

Citation Recall (50% of final score).

$$\text{tier1_recall} = |\text{tier1_gold} \cap \text{pred_set}| / |\text{tier1_gold}|$$

$$\text{tier2_recall} = |\text{tier2_gold} \cap \text{pred_set}| / |\text{tier2_gold}|$$

$$\text{weighted_recall} = 0.8 \times \text{tier1_recall} + 0.2 \times \text{tier2_recall}$$

$$\text{citation_score} = \text{weighted_recall} \times 100$$

Design rationale: Use Recall (not F1) because introduction writing is open-ended; examinees may reasonably cite works outside the gold set. Tier 1 (80% weight) represents indispensable core references; Tier 2 (20% weight) represents quality supplementary references. Tier 3 not scored.

Writing Rubric (50% of final score). LLM-as-Judge evaluates the introduction against paper-specific rubric dimensions. Each dimension scored 0–10, weighted by dimension weights, normalized to 0–100 scale.

Final Score.

$$\text{final_score} = 0.5 \times \text{citation_score} + 0.5 \times \text{rubric_score}$$

H.5 Model Generation (MG)

H.5.1 Task Overview. Model Generation (MG) is a Level 2 (comprehensive capability) task that evaluates models' ability to construct economic models from natural language scenarios. The core evaluation goal is to test whether models can clearly specify economic agents, optimization problems, constraints, and equilibrium/game mechanisms. Each paper produces 1 evaluation item containing a desymbolized scenario narrative, standard model structure, and element-counting rubric. The task depends only on PDF input and uses a 3-round generation pipeline.

H.5.2 Generation Workflow. MG employs a three-round pipeline to transform formal models into natural language scenarios:

- (1) **Round 1 (PDF input):** Fully extract the paper's core economic model, including all agents and their choice variables, objective functions (with optimization direction), all constraints (budget, resource, IC, PC, feasibility), equilibrium concept, key FOCs, and core mechanisms. For empirical papers, extract the theoretical framework or structural model if present.
- (2) **Round 2 (PDF + Round 1):** Rewrite Round 1's formal model into a desymbolized natural language scenario description. Remove all mathematical notation and variable symbols, replacing them with plain economic intuition and verbal descriptions of relationships.
- (3) **Round 3 (text input):** Construct an element-counting rubric based on Round 1's model structure. Dynamically allocate points across dimensions (agents, objectives, constraints, equilibrium characterization) based on the actual number of elements in the model.

H.5.3 Design Rationale. Round 1 focuses purely on faithful extraction without rewriting. Round 2 can then generate targeted natural language descriptions while referring back to the PDF to ensure economic intuition accuracy. Round 3 is purely computational—allocating rubric points based on element counts—and requires no PDF access.

H.5.4 Round 1 Prompt Template.

You are an economic theory expert. Read the attached PDF paper and
fully extract the paper's core economic model.

Extraction requirements

Strictly extract the following from the paper as written (do not omit any agent or constraint):

1. **Economic agents:** All agents in the model and their choice variables
2. **Objective functions:** Each agent's optimization objective (including maximization/minimization direction)
3. **Constraints:** All constraints each agent faces (budget, resource, incentive compatibility, participation, feasibility, etc.)
4. **Equilibrium concept:** The equilibrium / solution concept used
5. **Key first-order conditions:** FOCs / characterization results as derived or stated
6. **Core mechanism:** Economic forces that drive the model's results

Handling non-theoretical papers

If the paper is empirical:

- Extract the theoretical framework or structural model it relies on (if any)
- Extract the economic logic behind the identification strategy
- If the paper has no theoretical model at all, state that the paper is unsuitable for the MG task

When `suitable\_for\_mg = false`:

- Provide `reason\_not\_suitable`, briefly explaining why
- Other model fields may be empty arrays / empty objects (do not fabricate a model)

H.5.5 Round 2 Prompt Template.

You are an economics textbook author. Below is the full model structure extracted from an economics paper. Rewrite it as a **desymbolized economic scenario narrative** for examinees to model from.

IMPORTANT: Check suitability first

If the Round 1 output has `suitable\_for\_mg = false`, output a minimal JSON indicating the paper is unsuitable and stop.

Desymbolization requirements

1. **Remove ALL mathematical symbols and variable names**: No x , y , p , w , α , β , etc.
2. **Use plain economic language**: "consumption quantity" not " x ", "wage rate" not " w "
3. **Describe relationships verbally**: "utility increases with consumption" not " $U(x)$ is increasing"
4. **Preserve economic intuition**: The scenario must convey the same economic trade-offs as the formal model
5. **Self-contained**: Examinee should NOT need the original paper

Scenario structure

1. **Economic context** (2-3 sentences): What is the economic environment?
2. **Agents and their goals** (1 sentence per agent): Who are the decision-makers and what do they want?
3. **Constraints and trade-offs** (2-3 sentences): What limits their choices?
4. **Equilibrium/outcome concept** (1 sentence): What constitutes a solution?

Examinee task description

After the scenario, provide clear instructions: "Based on the scenario above, construct a formal economic model that captures the key trade-offs and decision problems. Your model should specify: (1) all economic agents and their choice variables, (2) each agent's objective function, (3) all relevant constraints, (4) the equilibrium concept."

H.5.6 Round 3 Prompt Template.

You are an exam rubric designer. Based on the model structure extracted in Round 1, construct an **element-counting rubric** that dynamically allocates points based on the actual number of model elements.

Rubric construction algorithm

Total points: 100

1. **Agent identification** (15-25 points): Allocate based on number of agents. If 1 agent: 15 pts. If 2 agents: 20 pts. If 3+ agents: 25 pts.
2. **Objective specification** (20-30 points): Allocate based on complexity. Simple single-period utility: 20 pts. Multi-period or strategic objectives: 25-30 pts.
3. **Constraint completeness** (30-40 points): Allocate proportionally to number of constraints. Each constraint type worth 5-10 pts depending on complexity.

4. **Equilibrium characterization** (15–20 points): Standard equilibrium concept: 15 pts. Complex equilibrium (e.g., perfect Bayesian): 20 pts.

Quality levels for each element

- **correct** (full weight): Element present, correctly formalized, economically meaningful
- **substantial** (70% weight): Mostly correct with minor issues
- **partial** (40% weight): Attempted but significant issues
- **minimal** (15% weight): Vaguely referenced without proper formalization
- **missing** (0 points): Not present

H.5.7 Scoring Method. MG uses element-counting rubric (total 100 points) with LLM-assisted validation of element correctness. Points are dynamically allocated based on model complexity. For each rubric element, LLM judges quality level (correct/substantial/partial/minimal/missing) and awards corresponding percentage of per-element points. Total score is sum of all element scores.

H.6 Economic Analysis (EA)

H.6.1 Task Overview. Economic Analysis (EA) is a Level 2 (comprehensive capability) task that evaluates models' ability to write professional, multi-dimensional economic analysis given data results or model outputs. The core evaluation goal is to test whether models can produce substantive economic interpretations of empirical or theoretical results. Each paper produces 1–2 evaluation items (one per core result), each containing a purely factual data description, a reference analysis paragraph, applicable scoring dimensions, and examinee instructions. The task depends only on PDF input and uses a 2-round generation pipeline.

H.6.2 Generation Workflow. EA uses a two-round pipeline to separate data description from analytical writing:

- (1) **Round 1 (PDF input):** Scan all tables and figures in the paper, label their type and importance, and select 1–2 core results that best represent the paper's contribution (typically main regression tables or core theoretical result figures). For each selected result, extract a context paragraph (150–250 words covering research question, data, methodology, and what the result represents) and a purely factual data description (200–400 words) without any interpretation or analysis.
- (2) **Round 2 (PDF + Round 1):** For each result identified in Round 1, locate the corresponding analysis paragraph(s) in the paper's main text (400–600 words). Extract the author's original analysis as the gold standard. Identify which of the six evaluation dimensions apply to this specific result.

H.6.3 Design Rationale. The core challenge is cleanly separating "result description" from "analysis paragraph." Round 1 identifies core results and extracts pure data descriptions (no interpretation), then Round 2 locates the author's analysis of these results as the gold standard. Single-round generation tends to blur the boundary between data description and analytical content.

H.6.4 Round 1 Prompt Template.

You are an expert in empirical economics research. Read the attached PDF paper, identify the paper's **core empirical/theoretical results**, and extract a purely factual data description.

Task 1: Scan all results in the paper

List all Tables and Figures in the paper (including titles). Label the type and importance of each:

- Type: main_regression | robustness | descriptive | heterogeneity | mechanism | theory_result | other
- Importance: core | supporting | auxiliary

Task 2: Select core results

Choose **1–2 results** that best represent the paper's core contribution. Selection criteria:

- Prefer main regression tables (e.g., Table 1–3) or core theoretical-result figures
- Avoid robustness checks, descriptive statistics, and other auxiliary tables
- Selected results should allow substantive analysis

Task 3: Write a rich context paragraph AND extract a purely factual data description

Context paragraph (field: `context`, 150–250 words): For each selected result, write a comprehensive background paragraph that enables a reader who has NEVER seen the paper to understand the result. This paragraph MUST include ALL of the following:

- **Research question & motivation** (2–3 sentences)
- **Data & sample** (1–2 sentences)

- **Methodology overview** (2–3 sentences)
- **What the result represents** (1 sentence)

CRITICAL: The context must NOT contain any interpretation, mechanism discussion, or analytical conclusions.

Data description (field: `data\_description`, 200–400 words): Describe the data/numbers themselves in plain text:

- You may include: coefficient values, standard errors, significance levels, sample size, R-squared, direction of trends, etc.
- Do NOT include: causal interpretation, mechanism explanation, policy implications, links to theory

Handling theoretical papers

If the paper is purely theoretical:

- `selected\_results` should focus on core propositions, theorems, or comparative statics results
- `data\_description` should state the formal result in plain language (what the theorem says) without explaining why it holds
- `result\_type` should be `theoretical\_result`

H.6.5 Round 2 Prompt Template.

You are an expert in economics research methods. Re-read the attached PDF paper and complete the following tasks.

Task 1: Locate and extract a FOCUSED reference analysis

For each `selected\_result` from Round 1, find the passage(s) in the main text where the author **directly analyzes/interprets that result**.

CRITICAL length constraint: The reference_analysis must be

400–600 words. This means:

- Extract only the **most essential analytical paragraphs** that directly interpret the selected result
- Do NOT include extended discussions that cover other results, lengthy institutional background, or multi-page analyses
- If the author's discussion spans multiple pages, **select and condense** to the 400–600 word core
- Prefer the paragraphs immediately following the result presentation

Task 2: Tag applicable scoring dimensions

For each result, label which of the following 6 dimensions are **actually supported by that result**. Use EXACTLY these dimension IDs:

- `accuracy`: Can the result's numbers/patterns be meaningfully checked for accuracy?
- `economic\_interpretation`: Does the result support substantive economic interpretation (magnitude, heterogeneity)?
- `mechanism`: Does the result support discussion of economic mechanisms or causal channels?
- `theory\_connection`: Can the result be meaningfully connected to theoretical predictions or prior literature?
- `policy\_implication`: Does the result have implications for policy?
- `writing\_quality`: Always applicable -- assesses logical coherence and professional language

Task 3: Build the complete task

Integrate data descriptions from Round 1 and reference analysis from this round into the final evaluation item.

H.6.6 Evaluation Dimensions. Six dimensions are used consistently across Round 2 annotation and scoring prompts:

- **Accuracy** (accuracy): Correctness of numerical descriptions, trends, and statistical claims
- **Economic interpretation** (economic_interpretation): Substantive interpretation of magnitudes, heterogeneity, or practical significance
- **Mechanism** (mechanism): Explanation of economic mechanisms or causal logic behind results

- **Theory connection** (theory_connection): Linkage to theoretical predictions or prior literature
- **Policy implication** (policy_implication): Discussion of policy relevance (only when results support it)
- **Writing quality** (writing_quality): Logical clarity and professional language

Not every dimension applies to every result. `writing_quality` is always applicable; `accuracy` is almost always applicable. For theoretical results, `accuracy` refers to whether the formal statement is correctly characterized, and `mechanism` refers to the economic intuition behind the result.

H.6.7 Examinee Input. Examinees receive: (1) a context paragraph describing the research question, data, and methodology (150–250 words), and (2) a purely factual data description (e.g., “Table 3 shows that the coefficient on X is -0.37 (SE = 0.12), significant at the 1% level”) without any analytical interpretation. They are asked to write a 300–500 word analysis paragraph interpreting the result across applicable dimensions.

H.6.8 Scoring Method. EA uses multi-dimension LLM-as-Judge scoring (0–10 points per applicable dimension):

- (1) **Accuracy** (0–10): 9–10: all factual claims match data, no errors; 6–8: minor inaccuracies, core facts correct; 3–5: some factual errors affecting interpretation; 0–2: major factual errors.
- (2) **Economic interpretation** (0–10): 9–10: insightful interpretation with proper economic reasoning; 6–8: adequate interpretation, some depth lacking; 3–5: superficial, mostly restates numbers; 0–2: no meaningful interpretation.
- (3) **Mechanism** (0–10): 9–10: clear identification of driving mechanisms; 6–8: some mechanisms discussed but not fully developed; 3–5: vague or incomplete; 0–2: no mechanism identified.
- (4) **Theory connection** (0–10): 9–10: strong theoretical grounding with specific references; 6–8: some theoretical context; 3–5: weak or generic theory connection; 0–2: no theory connection.
- (5) **Policy implication** (0–10): 9–10: nuanced, well-grounded policy discussion; 6–8: reasonable policy points; 3–5: superficial; 0–2: no policy discussion.
- (6) **Writing quality** (0–10): 9–10: publication-ready prose, logical flow, precise language; 6–8: clear and organized, minor style issues; 3–5: readable but poorly organized; 0–2: disorganized or unclear.

Normalized score = total points / maximum possible points (counting only applicable dimensions, each worth 10 points max).

H.7 Summary

Table 10 summarizes the generation pipeline characteristics across all six tasks.

Task	Level	Rounds	Items/Paper	Scoring	Key Challenge
EK	L1	2	20 questions	Rule-based	Knowledge coverage balance
MM	L1	2	3 questions	LLM-as-Judge	Self-containment of questions
PA	L2	3	1 item	LLM-as-Judge	Clean abstract/body separation
LR	L2	4	1 item	Hybrid	Citation tier classification
MG	L2	3	1 item	LLM-as-Judge	Desymbolization fidelity
EA	L2	2	1–2 items	LLM-as-Judge	Data/analysis boundary

Table 10: Summary of task generation pipeline characteristics.

All six tasks employ multi-round generation pipelines with clear separation of concerns: early rounds perform exhaustive extraction or structural analysis with PDF access, while later rounds focus on question design, rubric construction, or validation using structured intermediate outputs. This design ensures high-quality, diverse evaluation items while maintaining consistency and reproducibility across the benchmark.